

UNIVERSITY OF UDINE

Department DPIA
Via delle scienze 206
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CIRCUITI E SISTEMI ELETTRONICI

Stefano Saggini
1 lezione

Books

Teoria

Microelettronica

- Autori Richard c. Jaeger travis n.blalock
- Mc GrawHill



Books

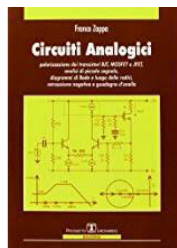
□ Eserciziari

□ Circuiti analogici

- Autore F. Zappa
- Progetto Leonardo

□ Esercizi di progettazione elettronica

- Autori Andrea Bonfanti
- Società editrice Esculapio



Program

□ Introduction

- Introduction analog design
- Analog components

□ Simple stages amplifier

- Frequency analysis method

□ Feedback loop

- Analysis method

□ Operational amplifier

- Introduction
- First stage
 - Differential Stage
 - Current mirrors
- Second stage
 - miller/ahuja compensation
- Output stages

□ Non linear circuits

- Bistable circuits and Oscillators

Introduzione Segnali

Segnale analogico e digitale

- *Segnale* è considerato qualsiasi valore misurabile di tensione, corrente o carica che descrive lo stato di un sistema o una grandezza fisica
 - ▣ *Segnale analogico* è definito in un continuo di valori di ampiezza e di tempo
 - ▣ *Segnale analogico campionato* è definito in un continuo di valori di ampiezza e in un insieme discreto di valori di tempo
 - ▣ *Segnale digitale* è definito in un insieme di valori discreti di tempo e di ampiezza

Introduzione Progetto e Analisi

Progetto e analisi

- *Analisi* è il processo estraere da uno schema circuitale le sue proprietà
 - ▣ Il risultato è unico
 - ▣ È realizzabile attraverso metodologie che possono essere automatizzate
- *Progetto* è il processo che parte dalle proprietà che sono definite come specifiche e termina con una soluzione circuitale
 - ▣ Il risultato non è unico è necessario introdurre delle cifre di merito per misurare le prestazioni
 - ▣ Il processo è automatizzabile solo parzialmente

Introduzione Organizzazione gerarchica del progetto

Ogni “prodotto” o “componente” è descritto da due e schemi:

- ❑ Schema funzionale: descrizione che contiene le informazioni mirate all' utilizzo del componente (data-sheet, manuale della tecnologia)
- ❑ Schema di realizzazione: descrizione che contiene le informazioni su come è costruito il componente basata su descrizioni funzionali dei componenti a livello gerarchico inferiore

Esempio

LM3100 SIMPLE SWITCHER® Synchronous 1MHz 1.5A Step-Down Voltage Regulator

General Description

The LM3100 Synchronously Rectified Buck Converter features all functions needed to implement a highly efficient, cost effective buck regulator capable of supplying 1.5A to loads with voltages as low as 0.8V. Dual 40V N-Channel synchronous MOSFET switches allow for low external component thus reducing complexity and minimizing board space. The LM3100 is designed to work exceptionally well with ceramic and other very low ESR output capacitors. The Constant ON-Time (COT) regulation scheme requires no loop compensation, results in fast load transient response, and simplifies circuit implementation. Through the use of a unique design the regulator does not rely on output capacitor ESR for stability, as do most other COT regulators. The operating frequency remains nearly constant with line and load variations due to the inverse relationship between the input voltage and the on-time. The operating frequency can be externally programmed up to 1MHz. Protection features include V_{CC} under-voltage lockout, thermal shutdown and gate drive under-voltage lockout. The part is available in a thermally enhanced eTSSOP-20 package.

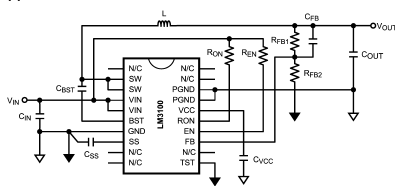
Features

- Input voltage range 4.5V - 36V
- 1.5A output current
- 0.8V, $\pm 1.5\%$, reference
- Integrated 40V, dual N-Channel buck synchronous switches
- Low component count and small solution size
- No loop compensation required
- Ultra-fast transient response
- Stable with ceramic and other low ESR capacitors
- Programmable switching frequency up to 1MHz
- Max. duty cycle limited during start-up
- Valley current limit
- Precision Internal Reference for adjustable output voltage down to 0.8V
- Thermal shutdown
- Thermally enhanced eTSSOP-20 package

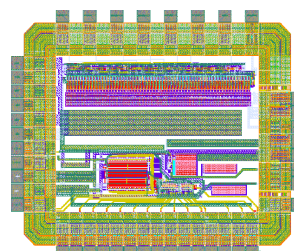
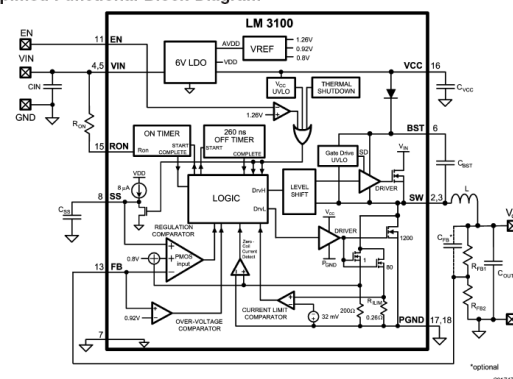
Typical Applications

- 5VDC, 12VDC, 24VDC, 12VAC, and 24VAC systems
- Embedded Systems
- Industrial Controls
- Automotive Telematics and Body Electronics
- Point of Load Regulators
- Storage Systems
- Broadband Infrastructure
- Direct Conversion from 2/3/4 Cell Lithium Batteries Systems

Typical Application

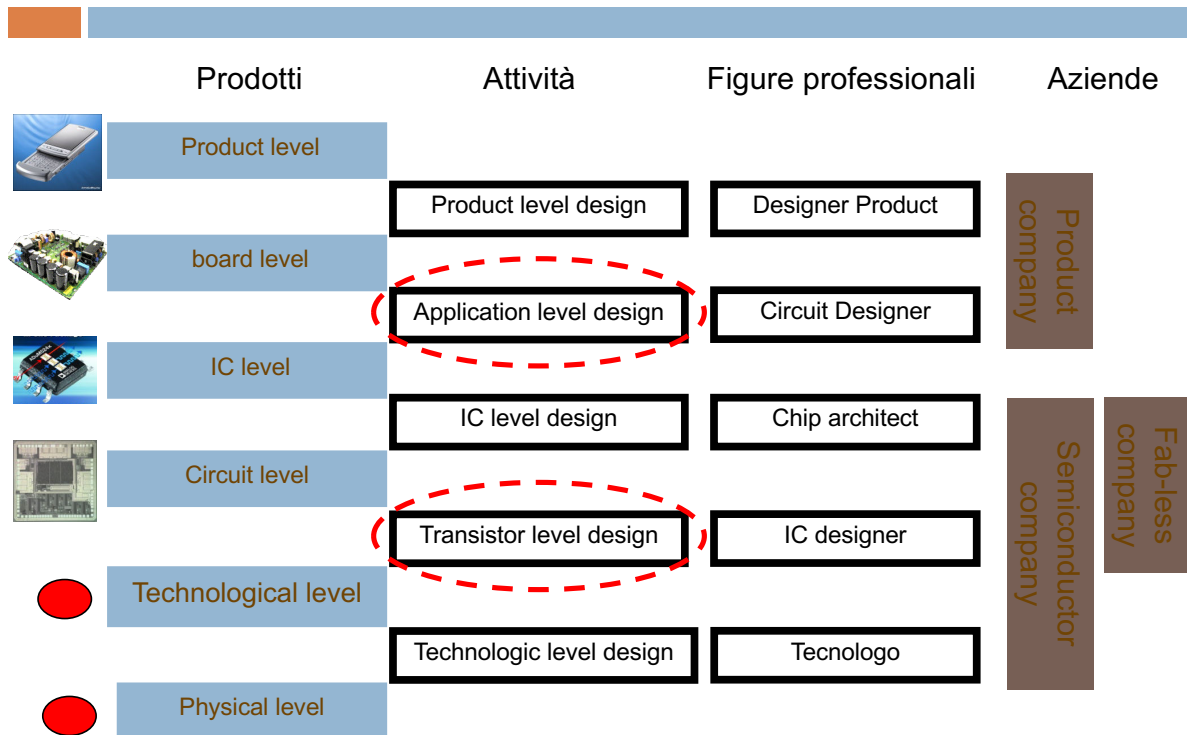


Simplified Functional Block Diagram



Introduzione

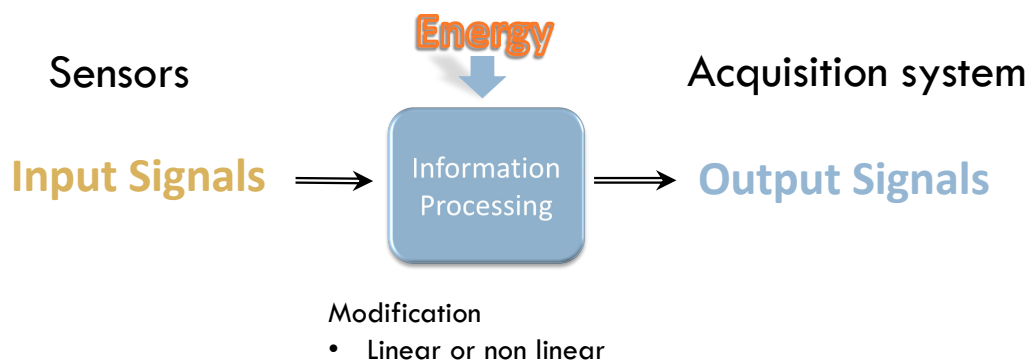
Organizzazione gerarchica del progetto



Example of design

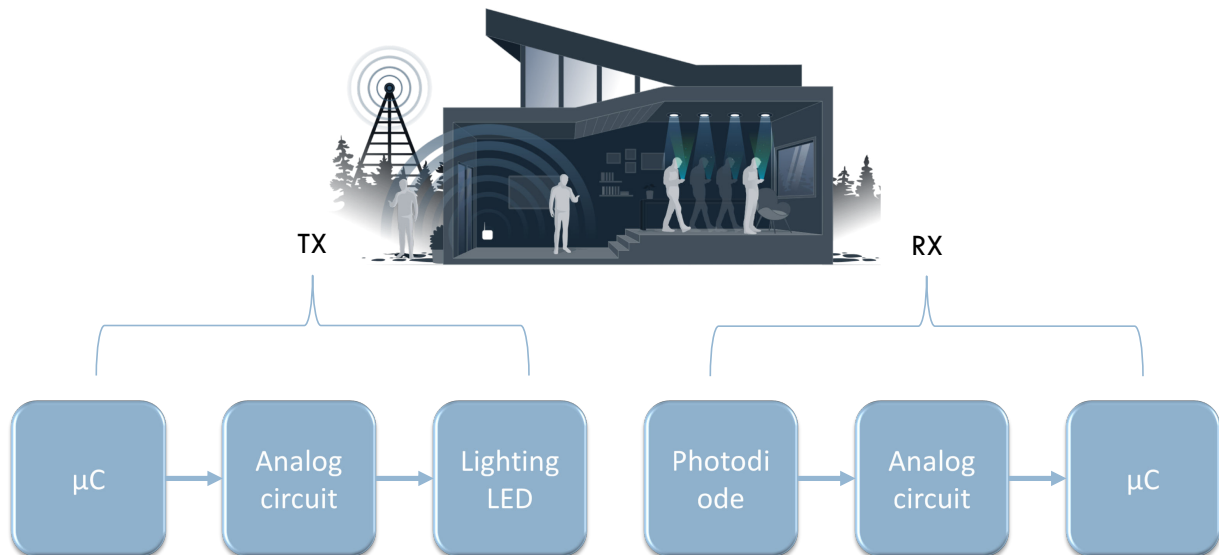
□ Discussing example related

▣ Analog processing



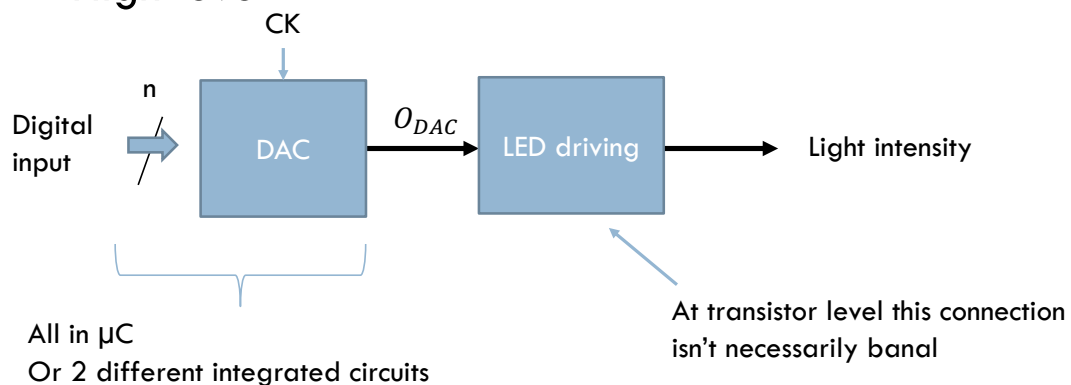
Example of design: LiFi

□ Specification



Design of TX

□ High level

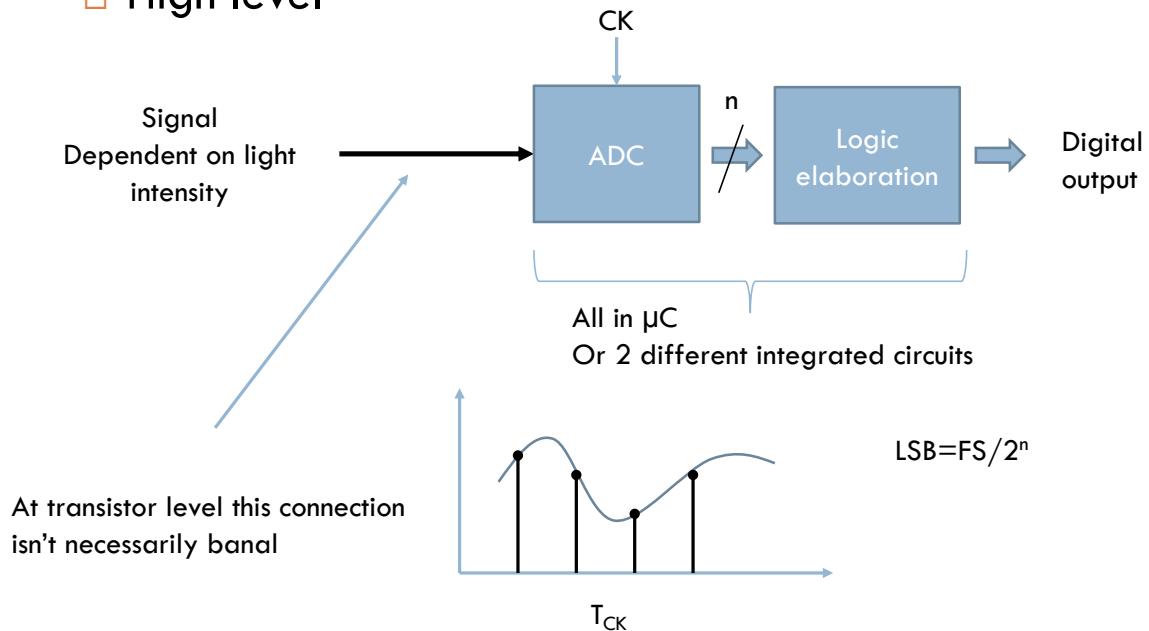


$$O_{DAC} = FS (b_1 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-n})$$

$$\text{Dove } b_i \in \{0,1\}$$

Design of RX

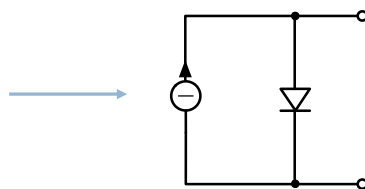
High level



Example of sensor: Photodiode

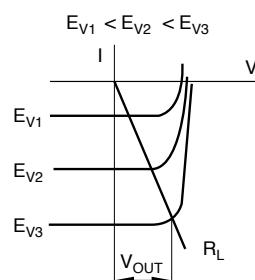
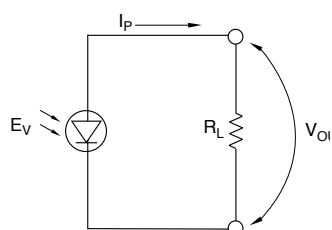
Sensor photodiode

Proportional to
The amount of
Incident light

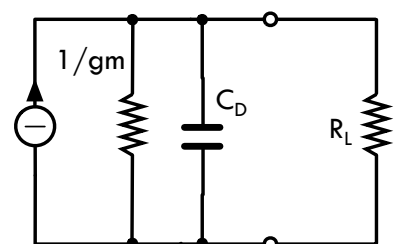


Large signal model

Large signal analysis



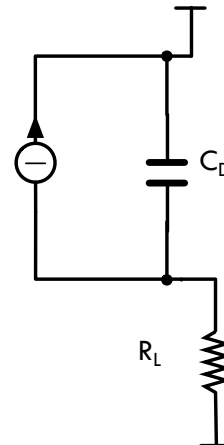
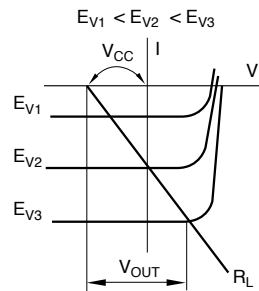
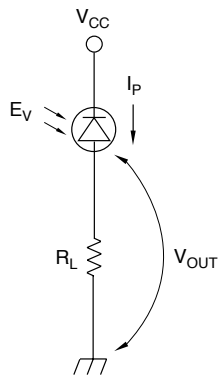
Small signal model



Example Photodiode

□ Polarized inverse

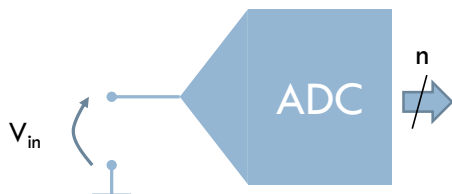
Small signal model



Basic ADC

□ ADC conversion two types

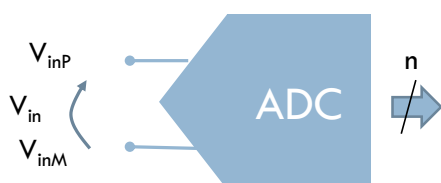
V_{in} not differential



V_{in} input conversion range
From a $V_{in\ MIN}$ to a $V_{in\ MAX}$ with n bit

Z_{in}  Input impedance

V_{in} differential



Each input has a dynamic range
from a $V_{P,M\ MIN}$ to a $V_{P,M\ MAX}$

The difference $V_{in} = V_{in\ P} - V_{in\ M}$
is converted From a $V_{in\ MIN}$ to a $V_{in\ MAX}$
with n bit

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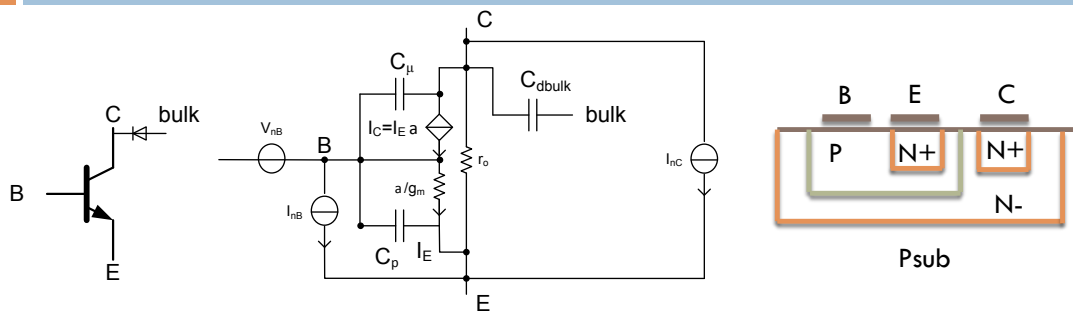
Stefano Saggini
2 lezione Tecnologie analogiche

Different Type of MOS available in the technology

- Core MOS (minimum)
 - ▣ Small minimum dimension L or W
 - ▣ small maximum V_{gs} , V_{gd} and V_d
 - ▣ Can be version Low V_{TH} or Zero V_{TH}
- I/O MOS (in general defined By the V_{dd})
 - ▣ Higher minimum dimension L or W
 - ▣ Higher maximum V_{gs} , V_{gd} and V_d
- High Voltage LDMOS
 - ▣ Fixed L dimension
 - ▣ High V_{ds} voltage



BJT



Small signal

$$gm = \frac{I_C}{V_T}$$

$$ro = \frac{V_E}{I_C}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

Approfondimento

Noise parameters

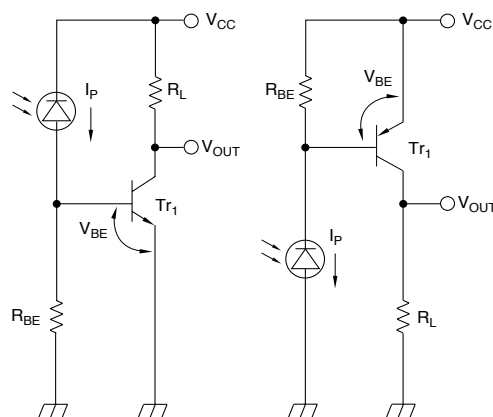
$$I_{nC} = 2qI_C$$

$$I_{nB} = 2qI_B$$

$$V_{nB} = 4kTR_b$$

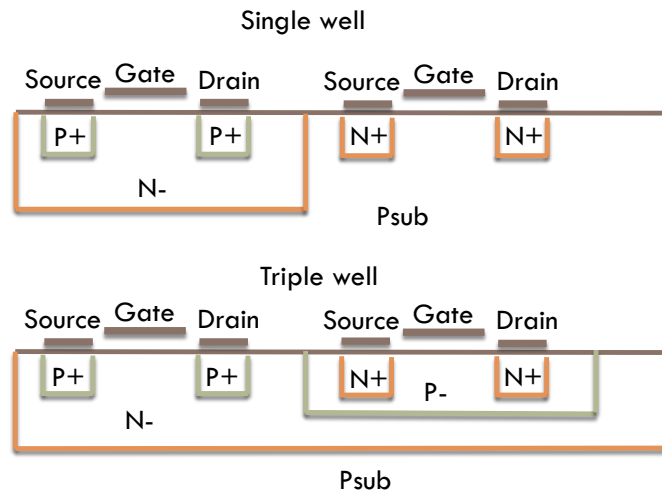
Our design first example

- Example of RX circuit with bipolar transistor



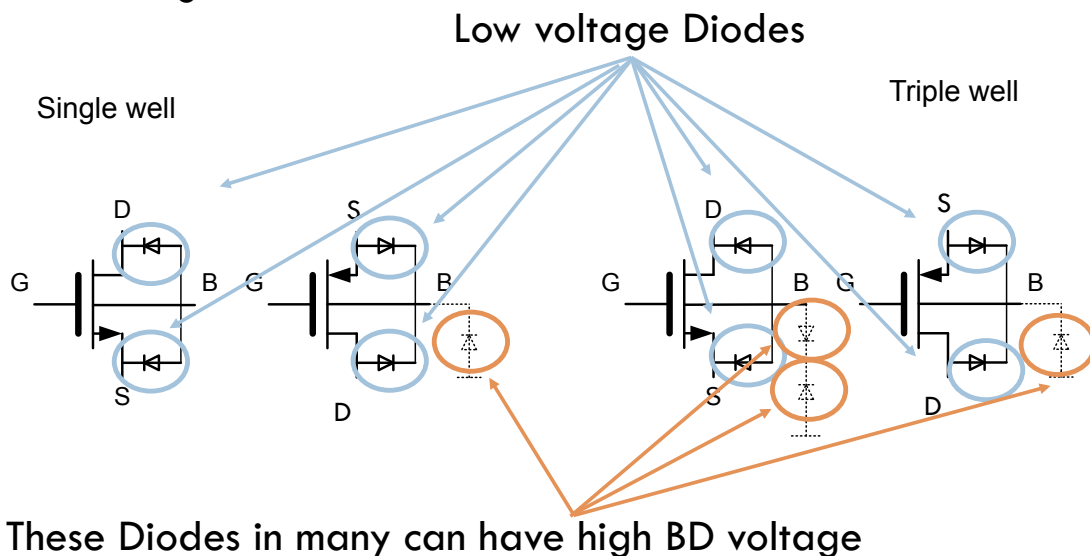
MOS model and description

- Different types of MOS available from the technology



MOS SOA consideration

- Voltage



MOS Description

□ MOS static model SPICE LEVEL I

TRIODE

$$I_D = \mu_o C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) - \left(\frac{V_{DS}}{2} \right) \right] V_{DS} \quad (V_{GS} - V_{TH}) \geq 0 \quad (V_{GD} - V_{TH}) \geq 0$$

SATURATION

$$I_D = \mu_o C_{ox} \frac{W}{2L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}) \quad (V_{GS} - V_{TH}) \geq 0 \quad (V_{GD} - V_{TH}) < 0$$

OFF STATE

$$I_D \cong 0 \quad (V_{GS} - V_{TH}) < 0 \quad (V_{GD} - V_{TH}) < 0$$

$$V_{TH} = V_{THO} + \gamma \left(\sqrt{2|\Phi_F| + V_{SB}} - \sqrt{2|\Phi_F|} \right)$$

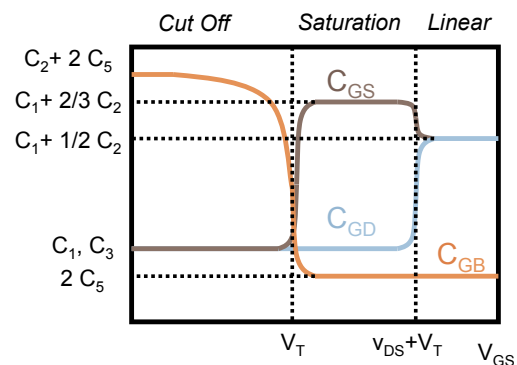
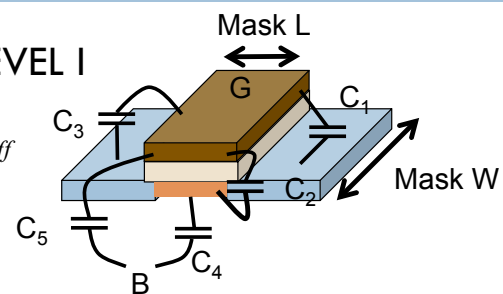
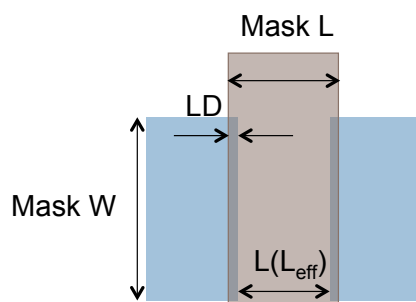
MOS Capacitors (AC)

□ MOS dynamic model SPICE LEVEL I

$$C_1 = C_3 \cong (LD) C_{ox} W_{eff} = (CGXO) W_{eff}$$

$$C_2 = (L - 2LD) W_{eff} C_{ox}$$

$$2C_5 = CGBO(L_{eff})$$



Short channel effects SCE

- Output resistance decreases due to channel length modulation.
- V_T becomes more dependent on L_{eff} and V_D .
- I_{off} and subthreshold slope increase; susceptibility to punch-through increases.
- Vertical and lateral fields increase.
- Mobility degrades; velocity saturates; hot-carrier effects become more important.
- Overlap and fringe capacitances become a larger fraction of gate capacitance.
- Extrinsic resistance becomes a larger fraction of source-to-drain resistance.
- Flicker noise and mismatch increase

Saturation velocity

- When the E field reach a critical value the carrier reach the saturation velocity

$$\tau \approx \frac{L_{eff}}{v_{sat}}$$

- Hence the current became

$$I_D \approx \frac{W_{eff} L_{eff} \bar{Q}}{\tau} \approx W_{eff} C_{ox} (V_{GS} - V_T) v_{sat}$$

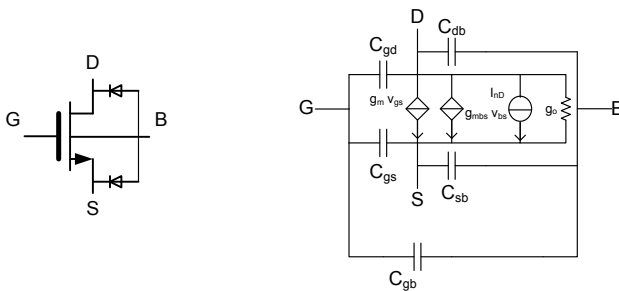
Average inversion charge for unit area

Small signal model in Saturation

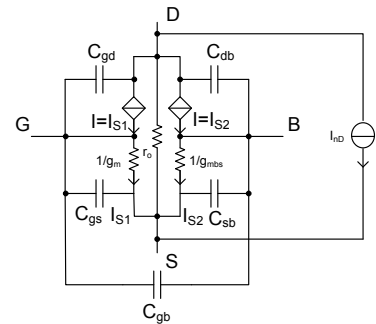
$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \sqrt{(2K'W/L)|I_D|(1+\lambda V_{DS})} = \sqrt{(2K'W/L)|I_D|}$$

$$g_{mbs} = -\frac{\partial I_D}{\partial V_{SB}} = -\frac{\partial I_D}{\partial V_T} \frac{\partial V_T}{\partial V_{SB}} = g_m \frac{\gamma}{2\sqrt{2|\Phi_F|+|V_{SB}|}} = \eta g_m$$

$$g_{ds} = g_o = \frac{\partial I_D}{\partial V_{SD}} = \frac{\lambda I_D}{1+\lambda V_{DS}} \approx \lambda I_D \quad (\lambda \downarrow L \uparrow)$$



L'impedenza di ingresso
del dispositivo è
rigorosamente un circuito
aperto in Continua!!!!!!!!!!!!!!



Under threshold MOS current

$$i_D \approx \frac{W}{L} I_{DO} e^{\frac{v_{GS}}{n}}$$

$1 < n < 3$ ← Process parameters

I_{DO} ←

Temperature dependence

Temperature

$$V_{TH}(T) = V_{TH}(T_o) + TCV(T - T_o)$$

$$\ln(K(T)) = \ln(K(T_o)) + BEX(\ln(T) - \ln(T_o))$$

$$TCV \cong -1.5 \text{ [mV/K]}$$

$$BEX \cong -1.76$$

In saturation the temperature effect depends on V_{ov} .
For low overdrive voltage the current increase with the temperature with high overdrive is the contrary

Noise generator

Drain current noise generator

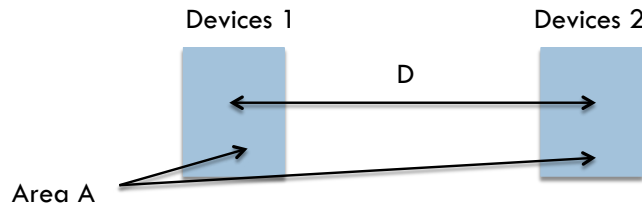
$$i_n^2 = \left[\frac{8k g_m (1 + \eta)}{3} + \frac{KF I_D}{2f C_{OX} L^2} \right] \Delta f \quad (A^2)$$

Reported at the input

$$e_n^2 = \frac{i_n^2}{g_m^2} = \left[\frac{8k(1 + \eta)}{3 g_m} + \frac{KF I_D}{2f C_{OX} W L K'} \right] \Delta f \quad (V^2)$$

Matching between two devices

- Considering two equal devices of area **A** on the same wafer characterized by the Parameter **P** (example the resistance, capacitor or the threshold voltage ecc..)



- P_1 and P_2 are two random variables
- The mean value of P on the wafer depend on process variation
- The difference $\Delta P = P_1 - P_2$ is Gaussian distributed with a variance

$$\sigma^2(\Delta P) = \frac{A_p^2}{A} + S_p^2 D^2$$

Experimentally determined

Matching between two devices

- In the technology manual is reported

- Absolute variation between parameters

$$\Delta P = P_1 - P_2$$

- Used for V_T of MOS

- Relative variation

$$\frac{\Delta P}{P} = \frac{200(P_1 - P_2)}{P_1 + P_2}$$

- Used for K of MOS, Resistance and capacitors

- Only the coefficient related to the Area **A**

Matching between two MOS

$$\Delta P = P_1 - P_2$$

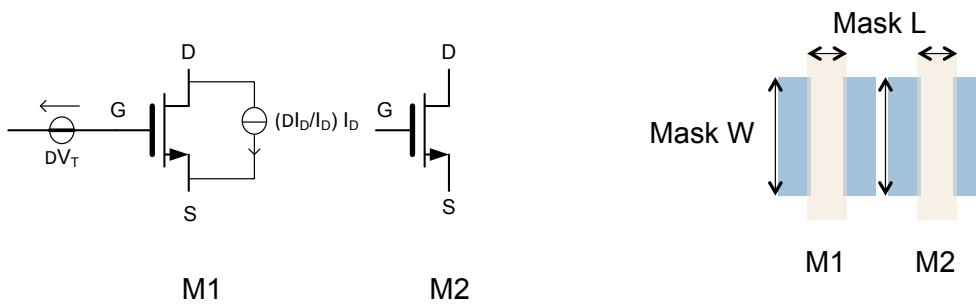
Used for V_T

$$\frac{\Delta P}{P} = \frac{200(P_1 - P_2)}{P_1 + P_2}$$

Used for K

$$\sigma(\Delta P) = \frac{\alpha}{\sqrt{WL}} \quad \sigma\left(\frac{\Delta P}{P}\right) = \frac{\beta}{\sqrt{WL}}$$

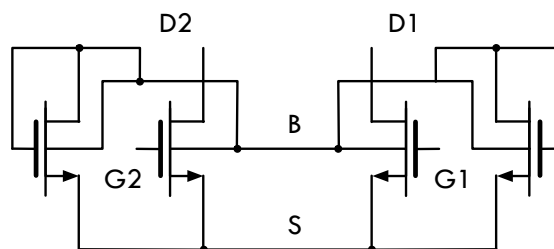
Standard deviation



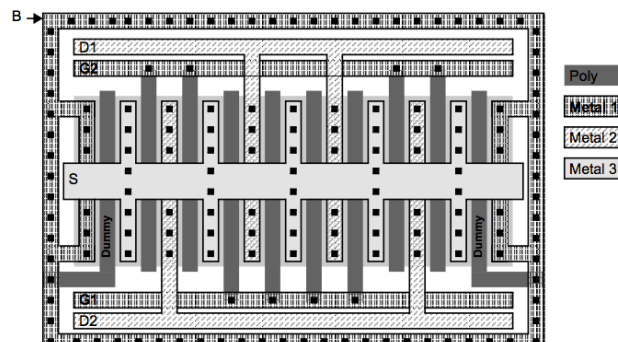
Recommended layout

Centroid

Schematic

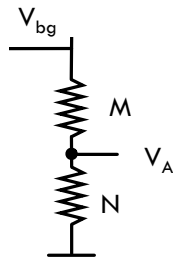


Layout



Examples for voltage dividers

□ Voltage references



$$V_A = \frac{M}{N+M} V_{bg}$$

Approfondimento

Calcolando la varianza della tensione otteniamo

$$\frac{\sigma V_A}{V_A} = \sqrt{\frac{M}{(N+M)N}} \frac{\sigma R}{R} \frac{1}{\sqrt{2}}$$

Scala con l'area

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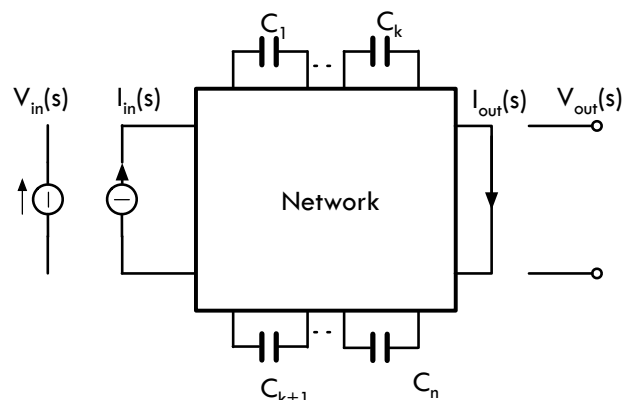


ANALISI IN FREQUENZA

Stefano Saggini
Complementi di circuiti e sistemi

Introduction

- Calculus of
 - ▣ Voltage or current gain
 - ▣ Impedance or transimpedance
 - ▣ Conductance or transconductance



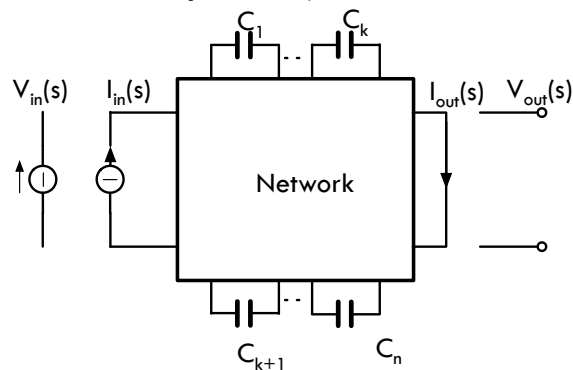
Soluzione del problema

- Calcolo della funzione di trasferimento tramite la soluzione del circuito lineare ottenuto sostituendo alle capacità la loro impedenza $Z_{Ck} = 1/(s C_k)$
- Tecnica semplificata che cerca di determinare la funzione di trasferimento attraverso le proprietà.

$$\frac{Y(s)}{U(s)} = G(0) \frac{\prod_{i=1}^{N_Z} (1 + s\tau_{zi})}{\prod_{j=1}^{N_P} (1 + s\tau_{pj})}$$

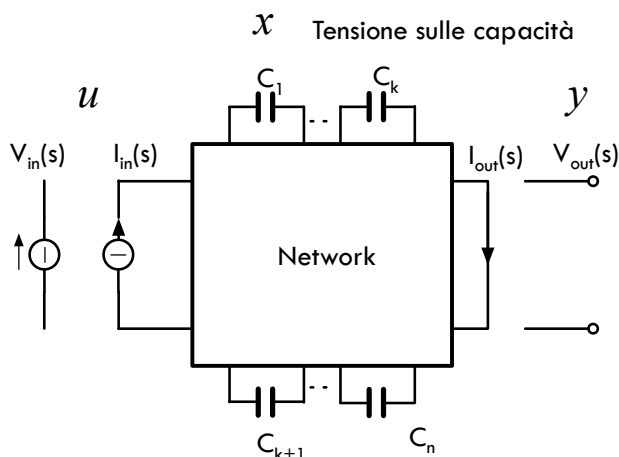
Determinazione di N_p

- Nel caso ci siano solo capacità nello schema il numero di poli è dato dal numero di **capacità indipendenti** (le loro tensioni non sono vincolate da generatori di tensioni pilotati e ne da equazioni di maglia con altre capacità)



Determinazione di N_z

- Se per $s \rightarrow \infty$ il trasferimento è $\neq$ da 0 allora $N_z = N_p$



x Tensione sulle capacità

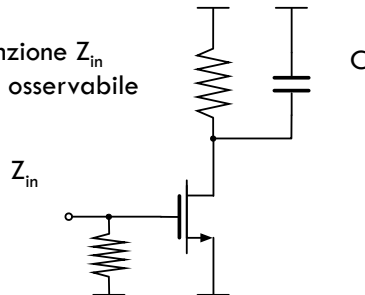
$$\begin{cases} \dot{x} = x A + B u \\ y = x C + D u \end{cases}$$

$$\frac{Y(s)}{X(s)} = C(sI - A)^{-1} B + D$$

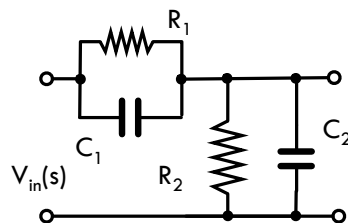
Cancellazione poli zeri

□ Caso di non osservabilità delle variabili di stato

Considerando la funzione Z_{in}
la capacità C è non osservabile



□ Caso di non raggiungibilità delle variabili di stato



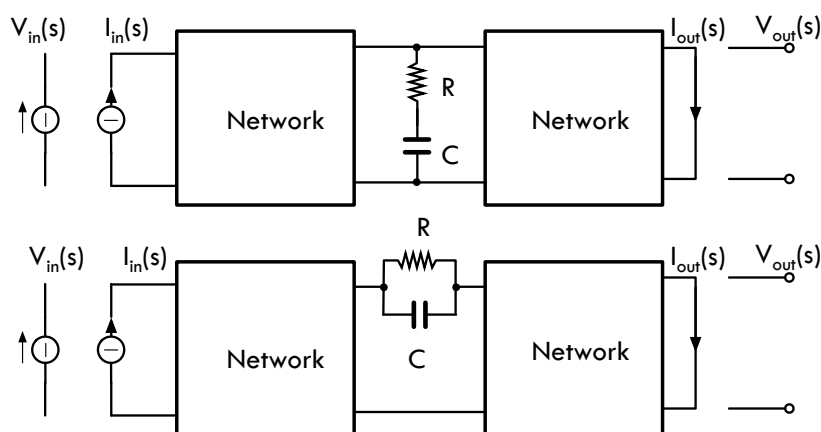
Considerando la funzione amplificazione di tensione
 A_v c'è solo una variabile di stato e se

$$\frac{C_2}{C_1} = \frac{R_1}{R_2}$$

La variabile è osservabile ma non raggiungibile

Determinazione di alcuni zeri

□ Certe configurazioni circuitali portano sempre degli zeri



$$\tau_z = RC$$

$$\tau_z = RC$$

Capacità interagenti

- Le capacità C_i e C_j sono interagenti se la tensione V_{C_i} influenza la corrente di carica I_{C_j} e viceversa.

$$\begin{cases} \dot{x} = x A + B u \\ y = x C + D u \end{cases}$$

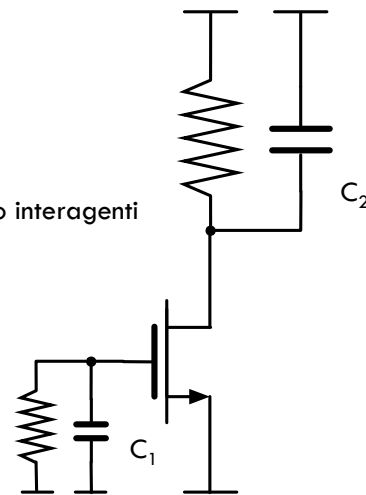
In altre parole i termini

$$a_{ij} \neq 0 \quad \text{e} \quad a_{ji} \neq 0$$

Esempio

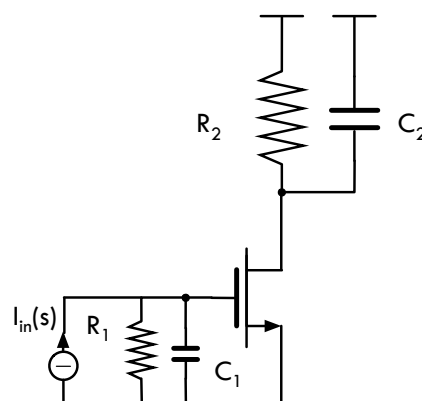
C_1 e C_2

Non sono interagenti



Proprietà

- I poli delle capacità non interagenti possono essere calcolate indipendentemente l'una dall'altra.



$$\tau_{p1} = R_1 C_1$$

$$\tau_{p2} = R_2 C_2$$

Metodo delle costanti di tempo

- Considerando una rete con capacità indipendenti e interagenti

$$\sum_{i=1}^{N_p} \tau_{pi} = \sum_{i=1}^{N_p} R'_i C_i$$

$$\sum_{i=1}^{N_p} \omega_{pi} = \sum_{i=1}^{N_p} \frac{1}{R''_i C_i}$$

Dove τ_i ($\omega_i = 1/\tau_i$) sono le costanti di tempo della rete e R'_i sono le resistenze calcolate ai morsetti di connessione delle capacità C_i con le altre capacità scollegate e R''_i sono le resistenze calcolate ai morsetti di connessione delle capacità C_i con le altre capacità cortocircuitate

Miller Theorem

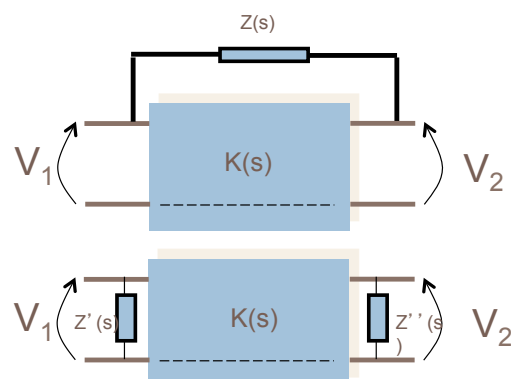
The impedance $Z(s)$ connected between V_1 e V_2 can be substituted with two impedances:

- $Z'(s)$ connected between V_1 and ground
- $Z''(s)$ connected between V_2 and ground

$$Z'(s) = \frac{Z(s)}{1 - K(s)}$$

$$Z''(s) = \frac{K(s)Z(s)}{1 - K(s)}$$

$$K(s) = \frac{V_2(s)}{V_1(s)}$$



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FEEDBACK SYSTEM 1

Associate professor Stefano Saggini
Lezione di complementi di elettronica

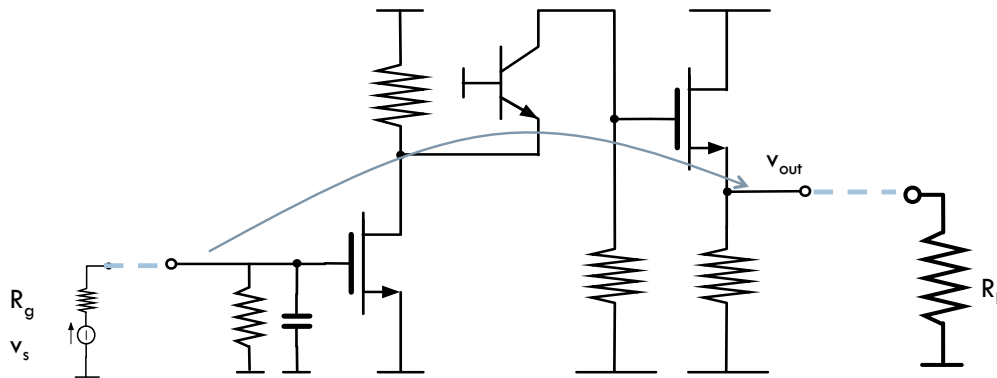
Sommario

- Introduzione ai sistemi multistadio
 - ▣ Rappresentazione come doppi bipoli
- Introduzione ai circuiti retroazionati
 - ▣ Calcolo del guadagno d'anello
 - ▣ Calcolo del guadagno ideale
 - ▣ Calcolo del guadagno diretto
 - ▣ Calcolo del guadagno reale

Amplificatore multistadio

□ Sistema multistadio

- Il Guadagno tra v_{out} e v_s dipende dai valori di R_L e R_g



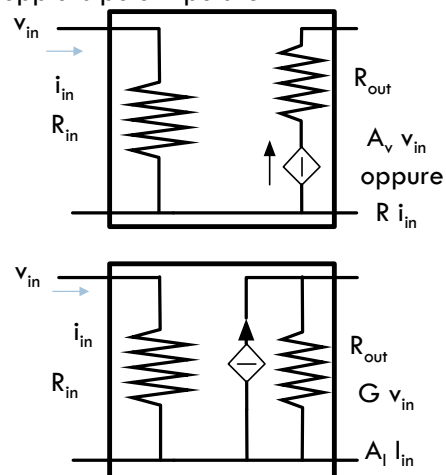
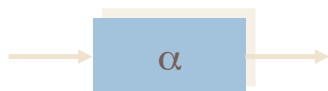
Descrizione dell'amplificatore

□ Circuito lineare per la descrizione dell'amplificatore

Questa rappresentazione è più adatta per il calcolo dell'effettivo guadagno:

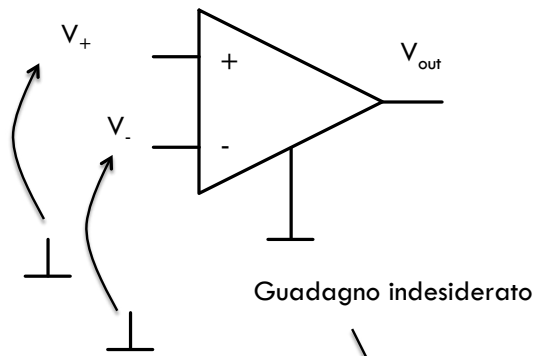
Doppio bipolo tripolare.

Questa rappresentazione non è sufficiente



Amplificatore differenziale

- Il segnale di ingresso differenziale in tensione



$$v_{DIF} = v_+ - v_-$$

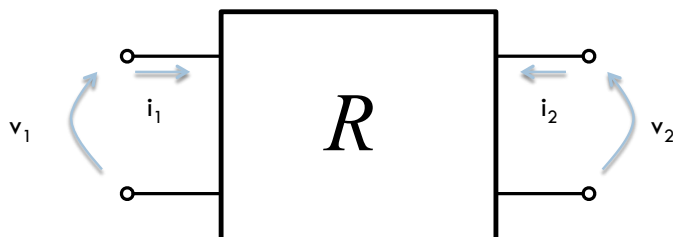
$$v_{CM} = \frac{v_+ + v_-}{2}$$

$$CMRR = \frac{A_D}{A_{CM}}$$

$$v_{OUT} = A_D v_{DIF} + A_{CM} v_{CM}$$

Rappresentazione matematica

- Rappresentazione matematica

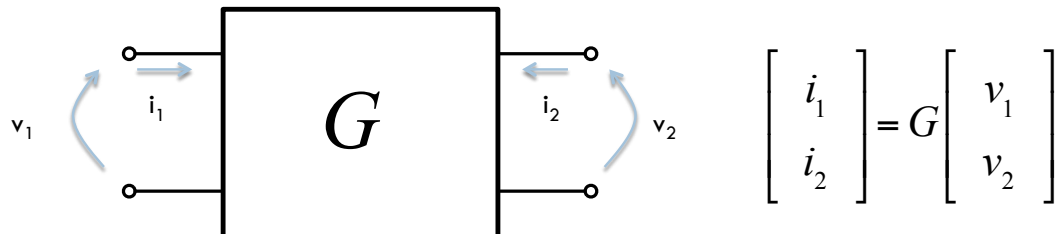


$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = R \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$R = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} = \begin{bmatrix} R_{in} & \cong 0 \\ R & R_{out} \end{bmatrix}$$

Rappresentazione matematica

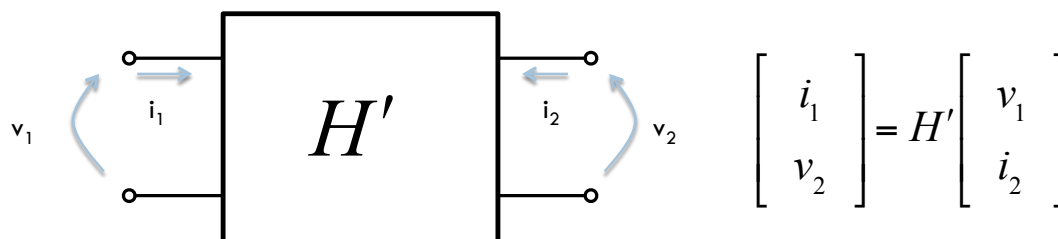
□ Rappresentazione matematica



$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} 1/R_{in} & \cong 0 \\ G & 1/R_{out} \end{bmatrix}$$

Rappresentazione matematica

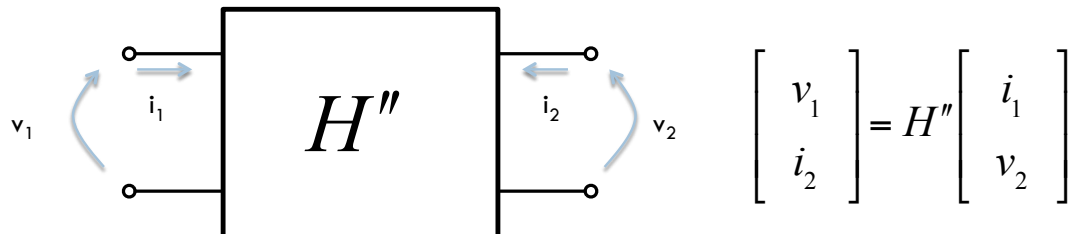
□ Rappresentazione matematica



$$H' = \begin{bmatrix} h'_{11} & h'_{12} \\ h'_{21} & h'_{22} \end{bmatrix} = \begin{bmatrix} 1/R_{in} & \cong 0 \\ A_V & R_{out} \end{bmatrix}$$

Rappresentazione matematica

□ Rappresentazione matematica



$$H'' = \begin{bmatrix} h''_{11} & h''_{12} \\ h''_{21} & h''_{22} \end{bmatrix} = \begin{bmatrix} R_{in} & \approx 0 \\ A_I & 1/R_{out} \end{bmatrix}$$

Circuiti retroazionati

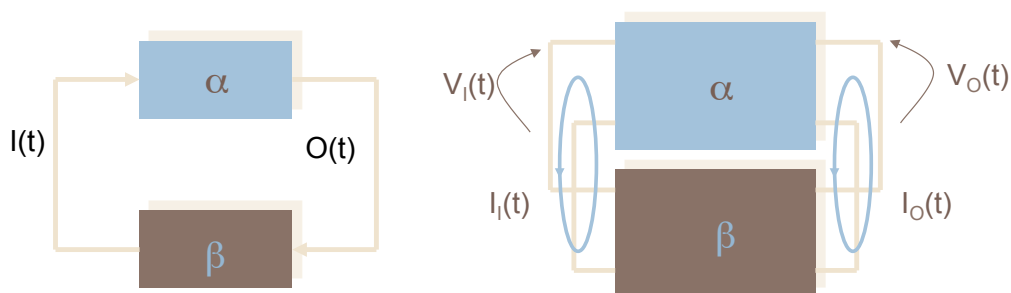
□ La retroazione:

- ▣ Stabilizza il guadagno rispetto alle caratteristiche parametriche dei componenti perché dipende dai componenti passivi
- Migliora la linearità

Teoria dei circuiti reazionati

Differenze tra lo schema di reazione ideale e il circuito con retroazione:

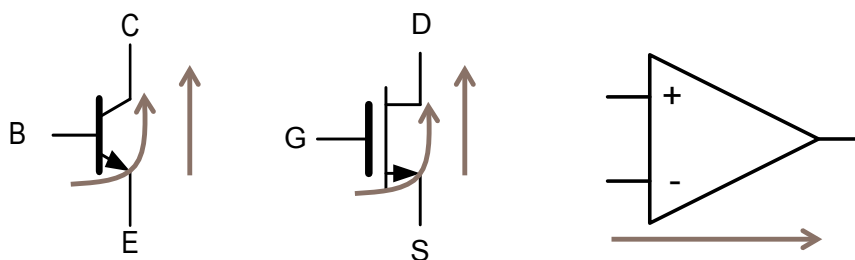
- Ogni blocco dello schema a blocchi ha una direzione e un trasferimento che non dipende dai blocchi a cui è collegato, lo schema elettrico non è sempre direzionale e il trasferimento dipende dagli elementi a cui è connesso
- I segnali dello schema elettrico possono essere di tensione o di corrente
- Non c'è una equivalenza netta tra lo schema a blocchi e lo schema circuitale c'è solo una similitudine, infatti il trasferimento complessivo non è ricavabile in modo immediato dal trasferimento degli schemi a blocchi.



Individuazione delle reazioni in uno schema

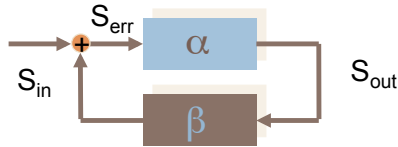
Per l'analisi della reazione è fondamentale individuare il blocco α e β :

- Per l'individuazione della direzione dell'anello è necessario seguire la direzione dei componenti direzionali di cui è composto lo schema

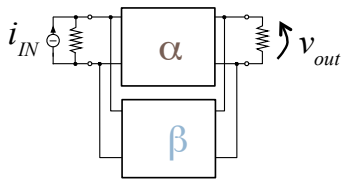


Reazione serie e parallelo

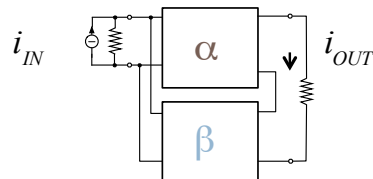
- The type of feedback (Voltage or current) depend on the connection



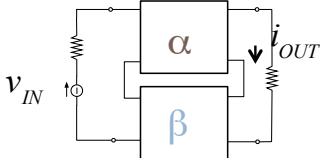
Input parallel the error is the current out is the voltage



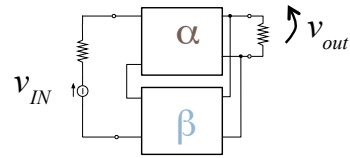
Input current connected in parallel, the error is a current. Output in series the output is the current.



Input in series the error is the voltage, out in series the current is the output.



Input in series the error is the voltage, out in parallel the voltage is the output.



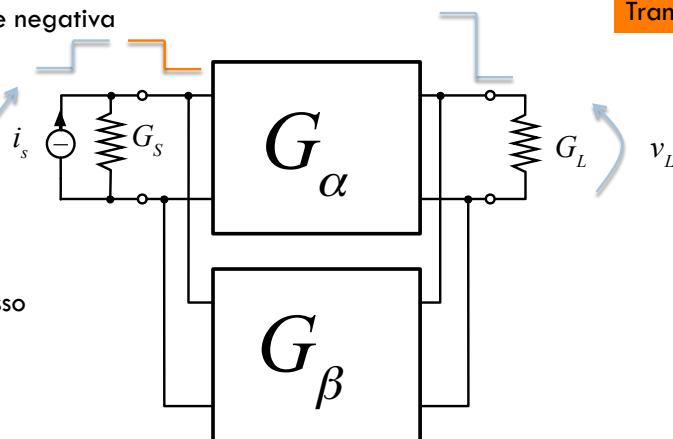
Retroazione parallelo parallelo

- Accesso ad un nodo della circuito retroazionato
l'uscita è su un nodo della retroazione

Effetto della
retroazione negativa

Intervento
dell'ingresso

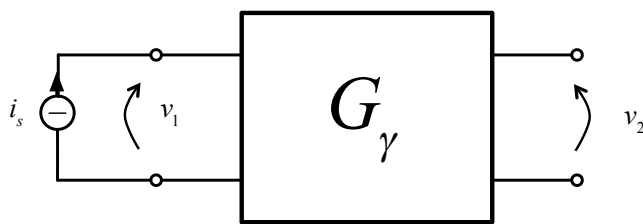
Transresistance



Retroazione parallelo parallelo

Calcolo del risultato

$$G_\gamma = G_\alpha + G_\beta + \begin{bmatrix} G_S & 0 \\ 0 & G_L \end{bmatrix} = \begin{bmatrix} g_{11}^\alpha + g_{11}^\beta + G_S & g_{12}^\alpha + g_{12}^\beta \\ g_{21}^\alpha + g_{21}^\beta & g_{22}^\alpha + g_{22}^\beta + G_L \end{bmatrix}$$



$$\begin{bmatrix} i_s \\ 0 \end{bmatrix} = G_\gamma \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$



$$G_\gamma^{-1} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Retroazione parallelo parallelo

Soluzione

$$\frac{1}{\det(G_\gamma)} \begin{bmatrix} g_{22}^\alpha + g_{22}^\beta + G_L & -(g_{12}^\alpha + g_{12}^\beta) \\ -(g_{21}^\alpha + g_{21}^\beta) & g_{11}^\alpha + g_{11}^\beta + G_S \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Analisi del primo trasferimento

$$\frac{v_2}{i_s} = \frac{-(g_{21}^\alpha + g_{21}^\beta)}{\det G_\gamma} = \frac{-g_{21}^\alpha}{\det G_\gamma} + \frac{-g_{21}^\beta}{\det G_\gamma}$$

Termine dominante R_D

Termine parassita R_P

Analisi del primo termine R_D

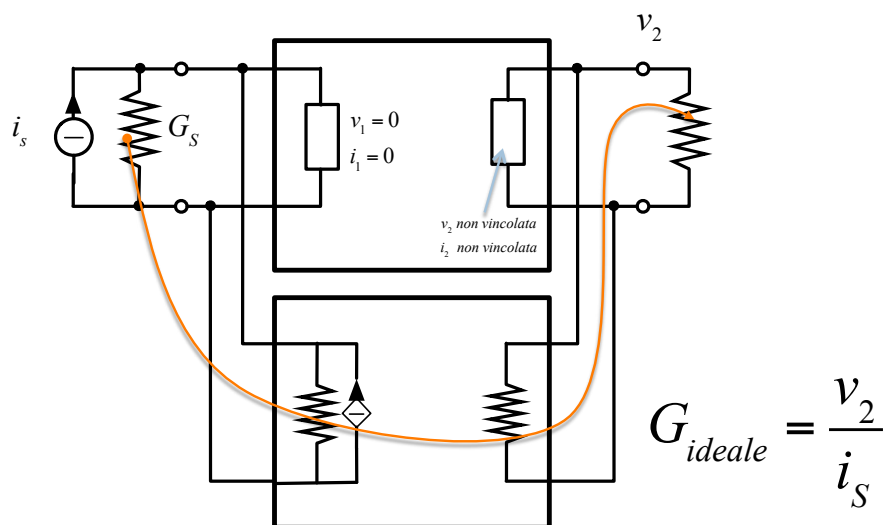
□ Analisi del termine R_D

$$R_D = \frac{-g_{21}^\alpha \quad \text{Guadagno di andata}}{(g_{11}^\alpha + g_{11}^\beta + G_S)(g_{22}^\alpha + g_{22}^\beta + G_L) - (g_{12}^\alpha + g_{12}^\beta)(g_{21}^\alpha + g_{21}^\beta)}$$

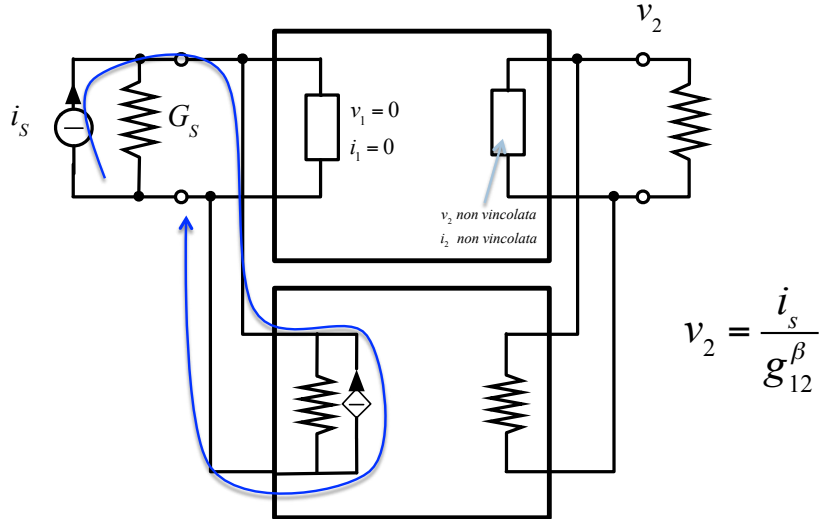
$$\lim_{G \rightarrow -\infty} R_D = \lim_{G \rightarrow -\infty} \frac{-G}{(g_{11}^\alpha + g_{11}^\beta + G_S)(g_{22}^\alpha + g_{22}^\beta + G_L) - (g_{12}^\alpha + g_{12}^\beta)(G + g_{21}^\beta)} = \frac{1}{(g_{12}^\alpha + g_{12}^\beta)} \equiv \frac{1}{g_{12}^\beta} = G_{ideale}$$

$$G_\alpha = \begin{bmatrix} 1/R_{in} & \approx 0 \\ G & 1/R_{out} \end{bmatrix}$$

G_{ideale} nel circuito



Dimostrazione



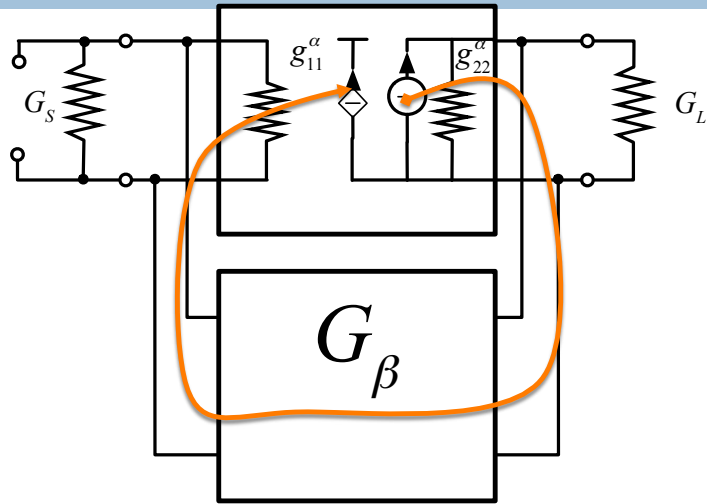
Analisi del primo termine R_D

$$R_D = \frac{1}{(g_{12}^{\alpha} + g_{12}^{\beta})} \frac{1}{\frac{(g_{21}^{\alpha} + g_{21}^{\beta})}{g_{21}^{\alpha}} - \frac{(g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}}$$

$$R_D = G_{ideale} \frac{1}{1 + \frac{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

$$G_{loop} = \frac{-(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

G_{loop} nel circuito



$$G_{loop} = \frac{-(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_S)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

Dimostrazione

$$G_{loop} = -g_{21}^{\alpha} \frac{v_1}{i_{test}}$$

$$\begin{bmatrix} 0 \\ i_{test} \end{bmatrix} = \begin{bmatrix} g_{11}^{\alpha} + g_{11}^{\beta} + G_S & g_{12}^{\alpha} + g_{12}^{\beta} \\ g_{21}^{\beta} & g_{22}^{\alpha} + g_{22}^{\beta} + G_L \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\frac{1}{(g_{11}^{\alpha} + g_{11}^{\beta} + G_S)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L) - g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta})} \begin{bmatrix} g_{22}^{\alpha} + g_{22}^{\beta} + G_L & -(g_{12}^{\alpha} + g_{12}^{\beta}) \\ -g_{21}^{\beta} & g_{11}^{\alpha} + g_{11}^{\beta} + G_S \end{bmatrix} \begin{bmatrix} 0 \\ i_{test} \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$v_1 = \frac{-(g_{12}^{\alpha} + g_{12}^{\beta})i_{test}}{(g_{11}^{\alpha} + g_{11}^{\beta} + G_S)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L) - g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta})}$$

$$G_{loop} = \frac{-(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_S)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

Analisi del secondo termine R_p

□ Analisi del termine R_p

$$R_p = \frac{-g_{21}^{\beta} \quad \text{Guadagno parassita}}{(g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L) - (g_{12}^{\alpha} + g_{12}^{\beta})(g_{21}^{\alpha} + g_{21}^{\beta})}$$

$$R_p = \frac{g_{21}^{\beta}}{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha} + (g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\beta} - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

$$R_p = \frac{g_{21}^{\beta}}{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha} - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)} \cdot \frac{1}{1 + \frac{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\alpha}}{(g_{12}^{\alpha} + g_{12}^{\beta})g_{21}^{\beta} - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}}$$

Analisi del secondo termine

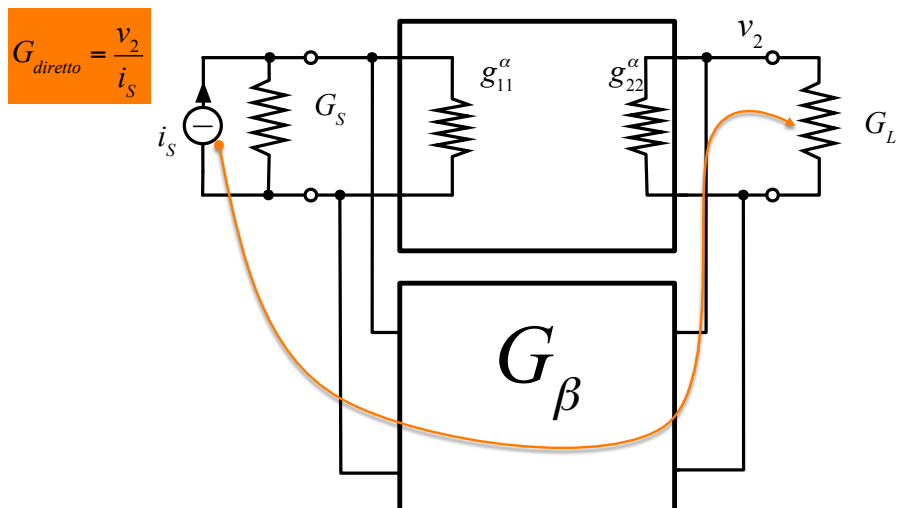
□ Analisi del secondo termine

$$R_p = \frac{g_{21}^{\beta}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)} \cdot \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{g_{21}^{\beta}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

$$R_p = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



Dimostrazione

$$G_{\text{diretto}} = \frac{v_2}{i_s}$$

$$\begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}^{\alpha} + g_{11}^{\beta} + G_s & g_{12}^{\alpha} + g_{12}^{\beta} \\ g_{21}^{\beta} & g_{22}^{\alpha} + g_{22}^{\beta} + G_L \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\frac{1}{(g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L) - g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta})} \begin{bmatrix} g_{22}^{\alpha} + g_{22}^{\beta} + G_L & -(g_{12}^{\alpha} + g_{12}^{\beta}) \\ -g_{21}^{\beta} & g_{11}^{\alpha} + g_{11}^{\beta} + G_s \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$v_2 = \frac{-g_{21}^{\beta} i_s}{(g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L) - g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta})}$$

$$\frac{v_2}{i_s} = G_{\text{diretto}} = \frac{g_{21}^{\beta}}{g_{21}^{\beta}(g_{12}^{\alpha} + g_{12}^{\beta}) - (g_{11}^{\alpha} + g_{11}^{\beta} + G_s)(g_{22}^{\alpha} + g_{22}^{\beta} + G_L)}$$

Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(G_\gamma)} \begin{bmatrix} g_{22}^\alpha + g_{22}^\beta + G_L & -(g_{12}^\alpha + g_{12}^\beta) \\ -(g_{21}^\alpha + g_{21}^\beta) & g_{11}^\alpha + g_{11}^\beta + G_S \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

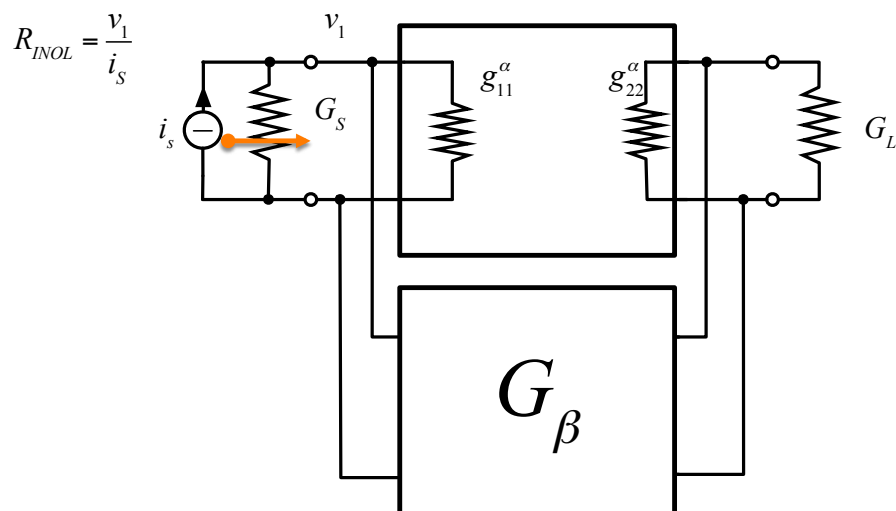
$$\frac{v_1}{i_s} = \frac{(g_{22}^\alpha + g_{22}^\beta + G_L)}{\det G_\gamma} = \frac{g_{22}^\alpha + g_{22}^\beta + G_L}{(g_{11}^\alpha + g_{11}^\beta + G_S)(g_{22}^\alpha + g_{22}^\beta + G_L) - (g_{12}^\alpha + g_{12}^\beta)(g_{21}^\alpha + g_{21}^\beta)}$$

$$R_{IN} = \frac{g_{22}^\alpha + g_{22}^\beta + G_L}{(g_{11}^\alpha + g_{11}^\beta + G_S)(g_{22}^\alpha + g_{22}^\beta + G_L) - (g_{12}^\alpha + g_{12}^\beta)g_{21}^\beta} \cdot \frac{1}{1 - \frac{(g_{12}^\alpha + g_{12}^\beta)g_{21}^\alpha}{(g_{11}^\alpha + g_{11}^\beta + G_S)(g_{22}^\alpha + g_{22}^\beta + G_L) - (g_{12}^\alpha + g_{12}^\beta)g_{21}^\beta}}$$

Resistenza anello aperto

$$R_{IN} = R_{INOL} \frac{1}{1 - G_{loop}}$$

$R_{\text{anello aperto}}$ nel circuito



Dimostrazione

$$R_{IN} = \frac{v_1}{i_s}$$

$$\begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}^\alpha + g_{11}^\beta + G_s & g_{12}^\alpha + g_{12}^\beta \\ g_{21}^\beta & g_{22}^\alpha + g_{22}^\beta + G_L \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\frac{1}{(g_{11}^\alpha + g_{11}^\beta + G_s)(g_{22}^\alpha + g_{22}^\beta + G_L) - g_{21}^\beta (g_{12}^\alpha + g_{12}^\beta)} \begin{bmatrix} g_{22}^\alpha + g_{22}^\beta + G_L & -(g_{12}^\alpha + g_{12}^\beta) \\ -g_{21}^\beta & g_{11}^\alpha + g_{11}^\beta + G_s \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$v_1 = \frac{(g_{22}^\alpha + g_{22}^\beta + G_L) i_s}{(g_{11}^\alpha + g_{11}^\beta + G_s)(g_{22}^\alpha + g_{22}^\beta + G_L) - g_{21}^\beta (g_{12}^\alpha + g_{12}^\beta)}$$

$$\frac{v_1}{i_s} = R_{INOL} = \frac{(g_{22}^\alpha + g_{22}^\beta + G_L)}{(g_{11}^\alpha + g_{11}^\beta + G_s)(g_{22}^\alpha + g_{22}^\beta + G_L) - g_{21}^\beta (g_{12}^\alpha + g_{12}^\beta)}$$

Trasferimento reale

Il trasferimento reale si può scomporre nei seguenti contributi di calcolo più immediato

$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

G_{reale} : Guadagno reale del trasferimento reazionato

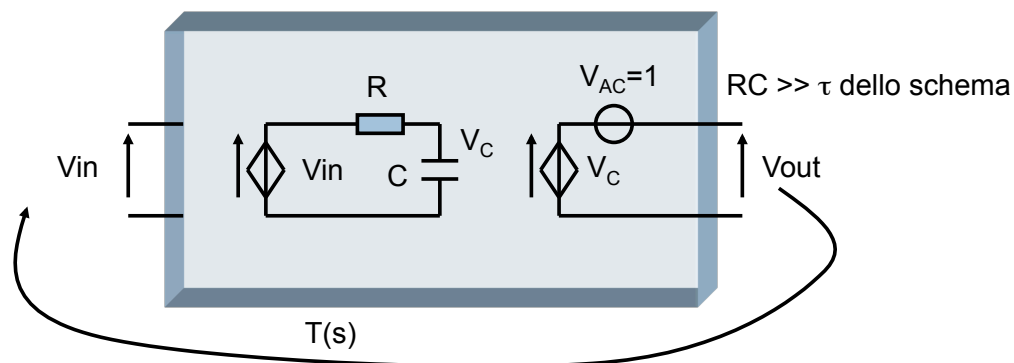
G_{loop} : Guadagno d'anello del circuito retroazionato

G_{ideale} : Guadagno reale del trasferimento se il G è infinito

$G_{diretto}$: Guadagno reale del trasferimento se il G è nullo

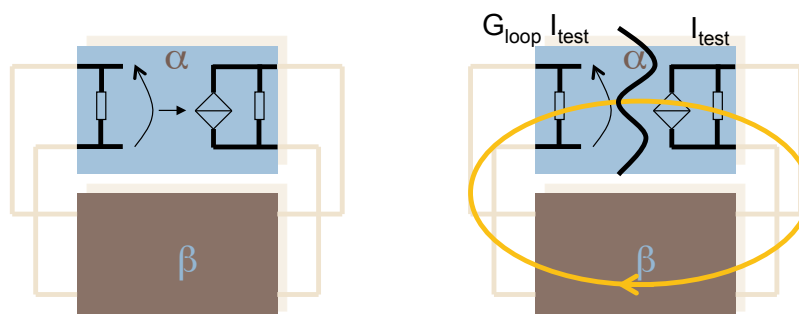
Calcolo del guadagno d'anello

- Il guadagno d'anello è il meccanismo che permette di ottenere il guadagno ideale
- Il calcolo del guadagno d'anello si ottiene spezzando l'anello in un punto "comodo" e calcolando il trasferimento sul circuito ottenuto ai capi del punto di rottura nella direzione dell'anello
- Si sopprimono gli ingressi (si aprono i generatori di corrente e si cortocircuitano quelli di tensione)
- Il calcolo si può anche realizzare in simulazione
- Questo calcolo è semplice da fare perché lo schema è direzionale



Calcolo del guadagno d'anello

Il punto di rottura più comodo è l'ingresso o l'uscita di una generatore comandato ideale altrimenti è necessario ricostruire le impedenze modificate dalla rottura



Calcolo del guadagno ideale

- Il guadagno ideale è un trasferimento che si ottiene facendo tendere a zero la variabile di ingresso della del circuito con retroazione, ovvero l'ingresso del blocco α sia essa corrente che tensione

Calcolo del guadagno diretto

Questo calcolo si ottiene annullando il guadagno del blocco α e calcolando il trasferimento tra ingresso ed uscita:

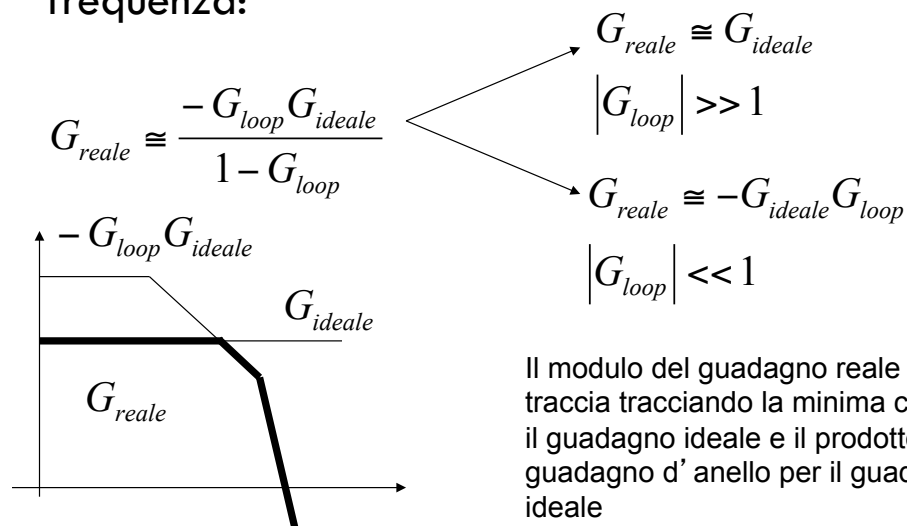
- Questo calcolo può essere realizzato in simulazione inserendo il blocco senza generatore di tensione AC utilizzato per il calcolo del guadagno d'anello connesso in modo da non perturbare il trasferimento diretto del segnale. Il generatore AC necessario per la valutazione di questo trasferimento va inserito all'ingresso del circuito.

Singularità del guadagno reale

- I poli del guadagno reale sono le soluzioni dell'equazione: $1 - G_{loop} = 0$
- Tutti i poli del guadagno ideale sono degli zeri del guadagno d'anello (non vale il viceversa)
- Il guadagno d'anello e il guadagno diretto hanno gli stessi poli

Rappresentazione del guadagno reale

- Rappresentazione del guadagno reale in frequenza:



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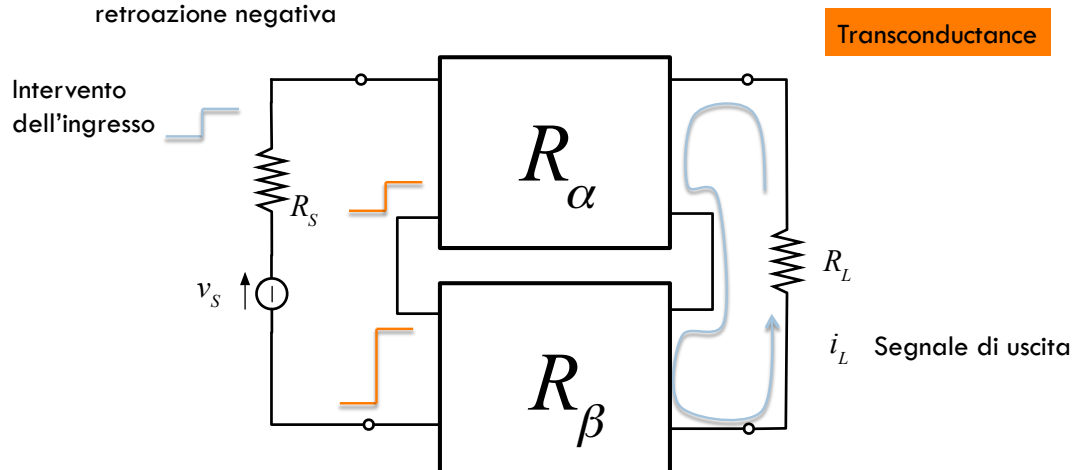
FEEDBACK SYSTEM 2

Associate professor Stefano Saggini
Lezione di complementi di elettronica

Retroazione serie serie

- Accesso ad un ramo della circuito retroazionato
l'uscita è su un ramo della retroazione

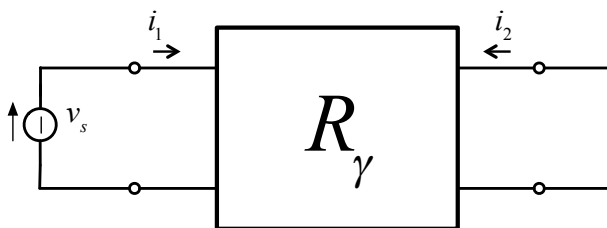
Effetto della
retroazione negativa



Retroazione serie serie

Calcolo del risultato

$$R_\gamma = R_\alpha + R_\beta + \begin{bmatrix} R_S & 0 \\ 0 & R_L \end{bmatrix} = \begin{bmatrix} r_{11}^\alpha + r_{11}^\beta + R_S & r_{12}^\alpha + r_{12}^\beta \\ r_{21}^\alpha + r_{21}^\beta & r_{22}^\alpha + r_{22}^\beta + R_L \end{bmatrix}$$



$$\begin{bmatrix} v_s \\ 0 \end{bmatrix} = R_\gamma \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$R_\gamma^{-1} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Retroazione serie serie

Soluzione

$$\frac{1}{\det(R_\gamma)} \begin{bmatrix} r_{22}^\alpha + r_{22}^\beta + R_L & -(r_{12}^\alpha + r_{12}^\beta) \\ -(r_{21}^\alpha + r_{21}^\beta) & r_{11}^\alpha + r_{11}^\beta + R_S \end{bmatrix} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Analisi del primo trasferimento

Termine dominante G_D

Termine parassita G_P

$$\frac{i_2}{v_s} = \frac{-(r_{21}^\alpha + r_{21}^\beta)}{\det R_\gamma} = \frac{-r_{21}^\alpha}{\det R_\gamma} + \frac{-r_{21}^\beta}{\det R_\gamma}$$

Trasferimento

Trasferimento

Termine dominante G_D

Termine parassita G_P

$$\frac{i_2}{v_s} = \frac{-(r_{21}^\alpha + r_{21}^\beta)}{\det R_\gamma} = \frac{-r_{21}^\alpha}{\det R_\gamma} + \frac{-r_{21}^\beta}{\det R_\gamma}$$

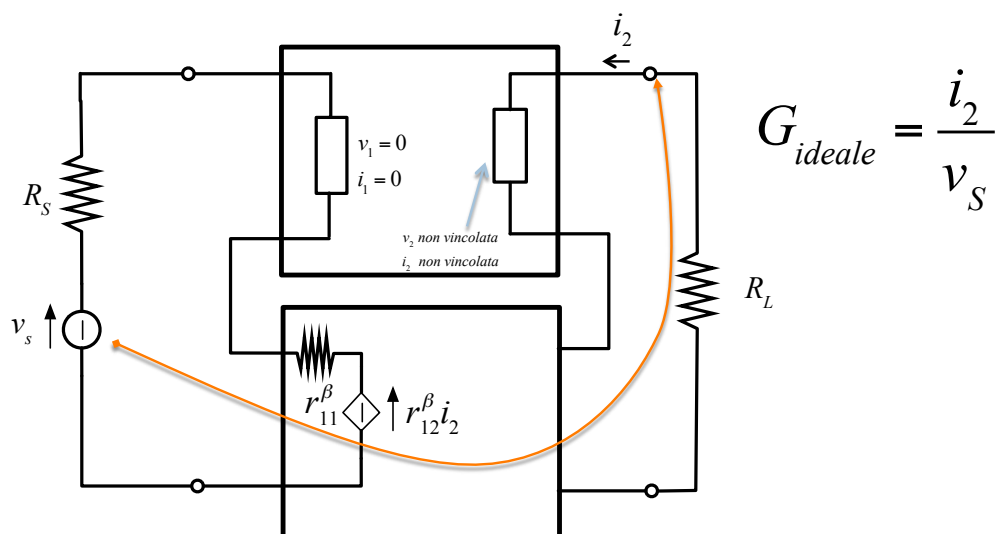
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$R_\alpha = \begin{bmatrix} R_{in} & \cong 0 \\ R & R_{out} \end{bmatrix}$$

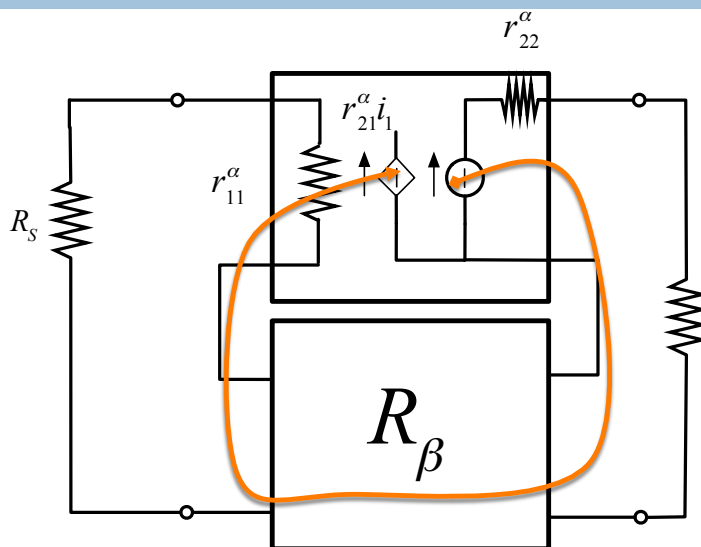
$$G_{ideale} = \lim_{R \rightarrow \infty} G_D = \lim_{R \rightarrow \infty} \frac{-R}{(r_{11}^\alpha + r_{11}^\beta + R_s)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)(R + r_{21}^\beta)} = \frac{1}{(r_{12}^\alpha + r_{12}^\beta)} \cong \frac{1}{r_{12}^\beta}$$

$$G_D = G_{ideale} \frac{1}{1 + \frac{r_{21}^\beta (r_{12}^\alpha + r_{12}^\beta) - (r_{11}^\alpha + r_{11}^\beta + R_s)(r_{22}^\alpha + r_{22}^\beta + R_L)}{(r_{12}^\alpha + r_{12}^\beta) r_{21}^\alpha}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



G_{loop} nel circuito



$$G_{loop} = \frac{-(r_{12}^{\alpha} + r_{12}^{\beta})r_{21}^{\alpha}}{r_{21}^{\beta}(r_{12}^{\alpha} + r_{12}^{\beta}) - (r_{11}^{\alpha} + r_{11}^{\beta} + R_S)(r_{22}^{\alpha} + r_{22}^{\beta} + R_L)}$$

Analisi del secondo termine

□ Analisi del secondo termine

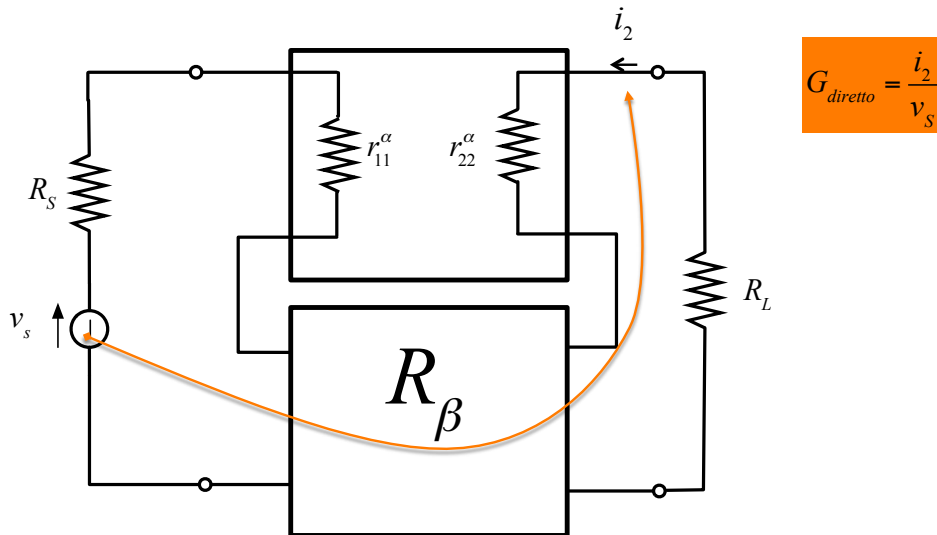
Guadagno diretto

$$G_P = \frac{r_{21}^{\beta}}{r_{21}^{\beta}(r_{12}^{\alpha} + r_{12}^{\beta}) - (r_{11}^{\alpha} + r_{11}^{\beta} + R_S)(r_{22}^{\alpha} + r_{22}^{\beta} + R_L)} \cdot \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{r_{21}^{\beta}}{r_{21}^{\beta}(r_{12}^{\alpha} + r_{12}^{\beta}) - (r_{11}^{\alpha} + r_{11}^{\beta} + R_S)(r_{22}^{\alpha} + r_{22}^{\beta} + R_L)}$$

$$G_P = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(R_\gamma)} \begin{bmatrix} r_{22}^\alpha + r_{22}^\beta + R_L & -(r_{12}^\alpha + r_{12}^\beta) \\ -(r_{21}^\alpha + r_{21}^\beta) & r_{11}^\alpha + r_{11}^\beta + R_s \end{bmatrix} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

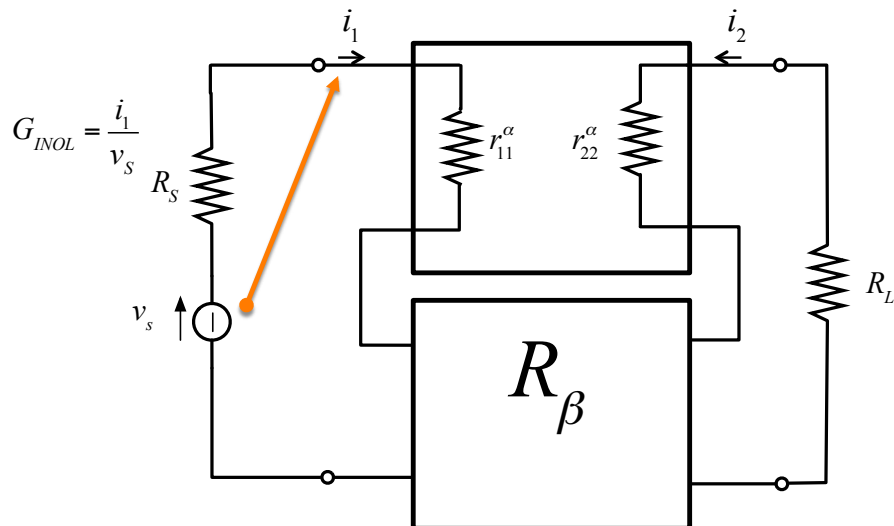
$$\frac{i_1}{v_s} = \frac{(r_{22}^\alpha + r_{22}^\beta + R_L)}{\det R_\gamma} = \frac{r_{22}^\alpha + r_{22}^\beta + R_L}{(r_{11}^\alpha + r_{11}^\beta + R_s)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)(r_{21}^\alpha + r_{21}^\beta)}$$

$$G_{IN} = \frac{r_{22}^\alpha + r_{22}^\beta + R_L}{(r_{11}^\alpha + r_{11}^\beta + R_s)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)r_{21}^\beta} \cdot \frac{1}{1 - \frac{(r_{12}^\alpha + r_{12}^\beta)r_{21}^\alpha}{(r_{11}^\alpha + r_{11}^\beta + R_s)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)r_{21}^\beta}}$$

Conduttanza anello aperto

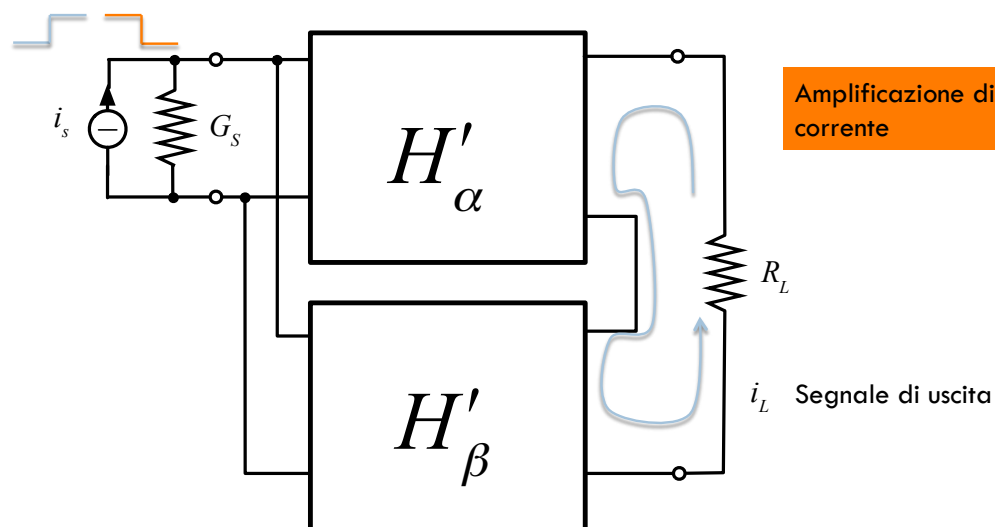
$$G_{IN} = G_{INOL} \frac{1}{1 - G_{loop}}$$

G_{anello aperto} nel circuito



Retroazione parallelo serie

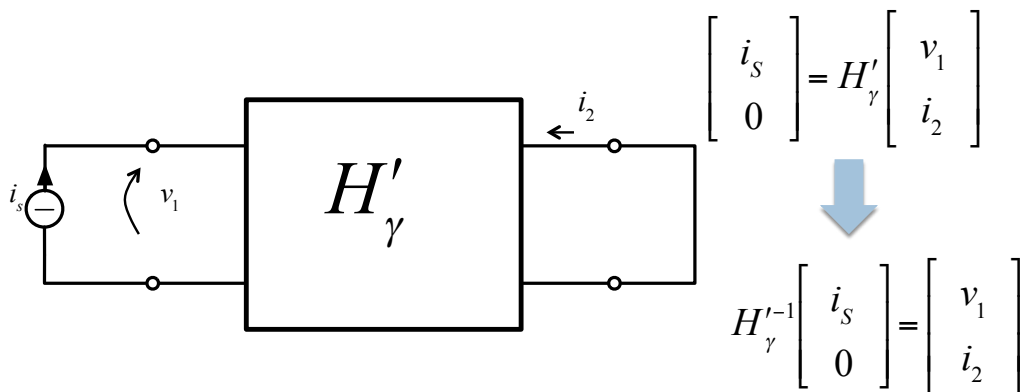
- Accesso ad un nodo del circuito retroazionato
l'uscita è su un ramo della retroazione



Retroazione parallelo parallelo

Calcolo del risultato

$$H'_\gamma = H'_\alpha + H'_\beta + \begin{bmatrix} G_S & 0 \\ 0 & R_L \end{bmatrix} = \begin{bmatrix} h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S & h'_{12}{}^\alpha + h'_{12}{}^\beta \\ h'_{21}{}^\alpha + h'_{21}{}^\beta & h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L \end{bmatrix}$$



Retroazione parallelo serie

Soluzione

$$\frac{1}{\det(H'_\gamma)} \begin{bmatrix} h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L & -(h'_{12}{}^\alpha + h'_{12}{}^\beta) \\ -(h'_{21}{}^\alpha + h'_{21}{}^\beta) & h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

Analisi del primo trasferimento

Termine dominante A_{ID}

Termine parassita A_{IP}

$$\frac{i_2}{i_s} = \frac{-(h'_{21}{}^\alpha + h'_{21}{}^\beta)}{\det H'_\gamma} = \frac{-h'_{21}{}^\alpha}{\det H'_\gamma} + \frac{-h'_{21}{}^\beta}{\det H'_\gamma}$$

Trasferimento

Trasferimento

Termine dominante A_{ID}

Termine parassita A_P

$$\frac{i_2}{i_S} = \frac{-(h'_{21}{}^{\alpha} + h'_{21}{}^{\beta})}{\det H'_{\gamma}} = \frac{-h'_{21}{}^{\alpha}}{\det H'_{\gamma}} + \frac{-h'_{21}{}^{\beta}}{\det H'_{\gamma}}$$

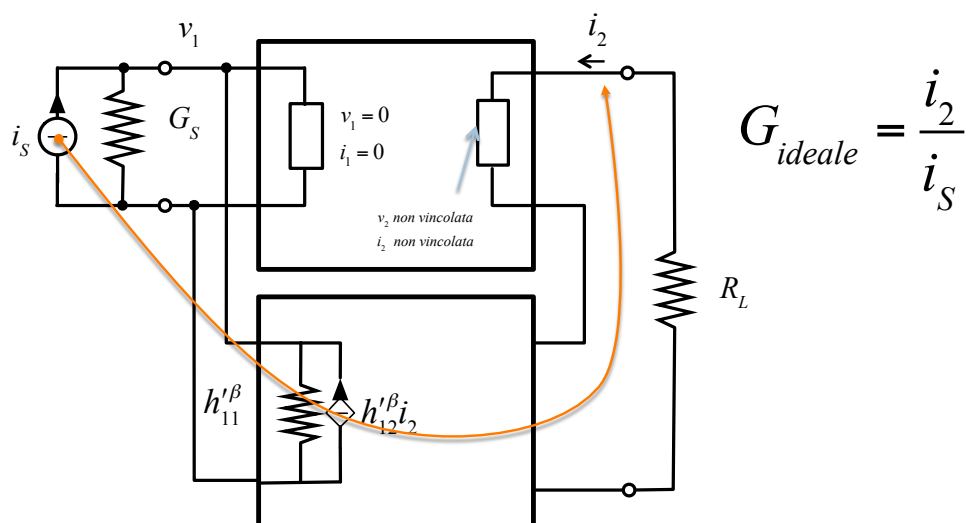
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$H'_{\alpha} = \begin{bmatrix} 1/R_{in} & \cong 0 \\ A_V & R_{out} \end{bmatrix}$$

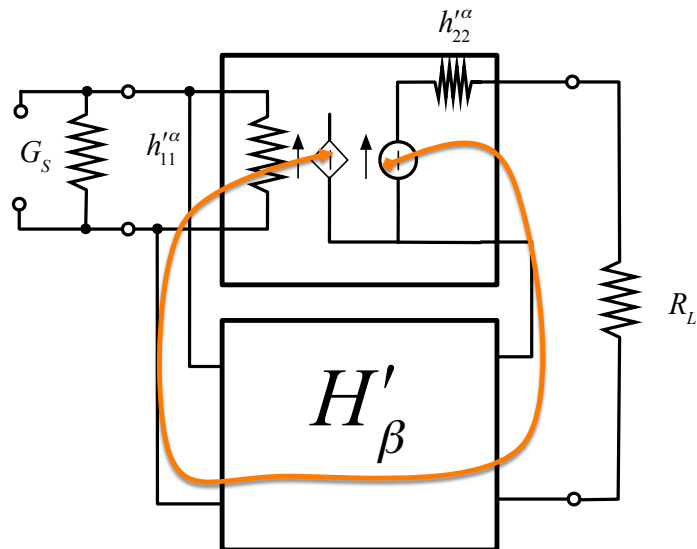
$$G_{ideale} = \lim_{A_V \rightarrow -\infty} A_{ID} = \lim_{A_V \rightarrow -\infty} \frac{-A_V}{(h'_{11}{}^{\alpha} + h'_{11}{}^{\beta} + G_S)(h'_{22}{}^{\alpha} + h'_{22}{}^{\beta} + R_L) - (h'_{12}{}^{\alpha} + h'_{12}{}^{\beta})(A_V + h'_{21}{}^{\beta})} = \frac{1}{(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta})} \cong \frac{1}{h'_{12}{}^{\beta}}$$

$$G_D = G_{ideale} \frac{1}{1 + \frac{h'_{21}{}^{\beta}(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta}) - (h'_{11}{}^{\alpha} + h'_{11}{}^{\beta} + G_S)(h'_{22}{}^{\alpha} + h'_{22}{}^{\beta} + R_L)}{(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta})h'_{21}{}^{\alpha}}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



G_{loop} nel circuito



$$G_{loop} = \frac{-(h'_{12} + h'_{12})h'_{21}}{h'_{21}(h'_{12} + h'_{12}) - (h'_{11} + h'_{11} + G_S)(h'_{22} + h'_{22} + R_L)}$$

Analisi del secondo termine

□ Analisi del secondo termine

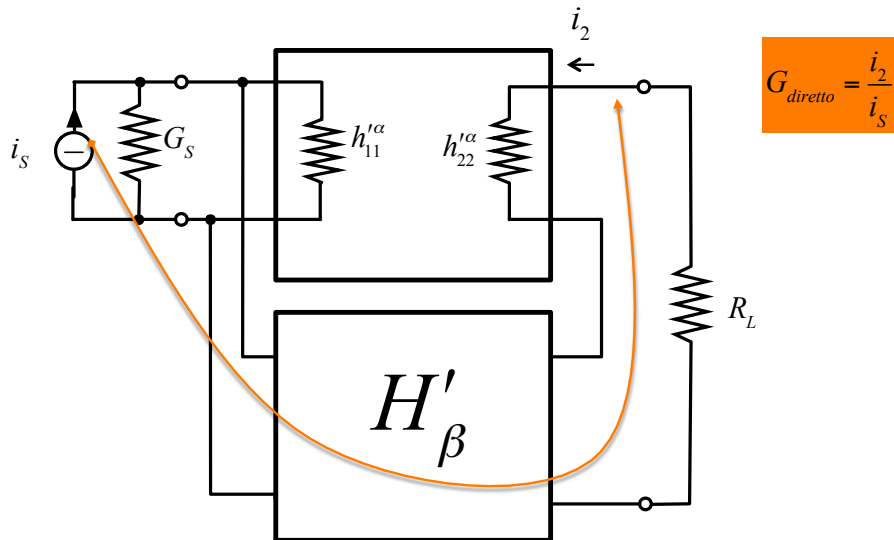
Guadagno diretto

$$A_{IP} = \frac{h'_{21}}{h'_{21}(h'_{12} + h'_{12}) - (h'_{11} + h'_{11} + G_S)(h'_{22} + h'_{22} + R_L)} \cdot \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{h'_{21}}{h'_{21}(h'_{12} + h'_{12}) - (h'_{11} + h'_{11} + G_S)(h'_{22} + h'_{22} + R_L)}$$

$$G_P = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(H'_\gamma)} \begin{bmatrix} h'_{22} + h'_{22} + R_L & -(h'_{12} + h'_{12}) \\ -(h'_{21} + h'_{21}) & h'_{11} + h'_{11} + G_s \end{bmatrix} \begin{bmatrix} i_s \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

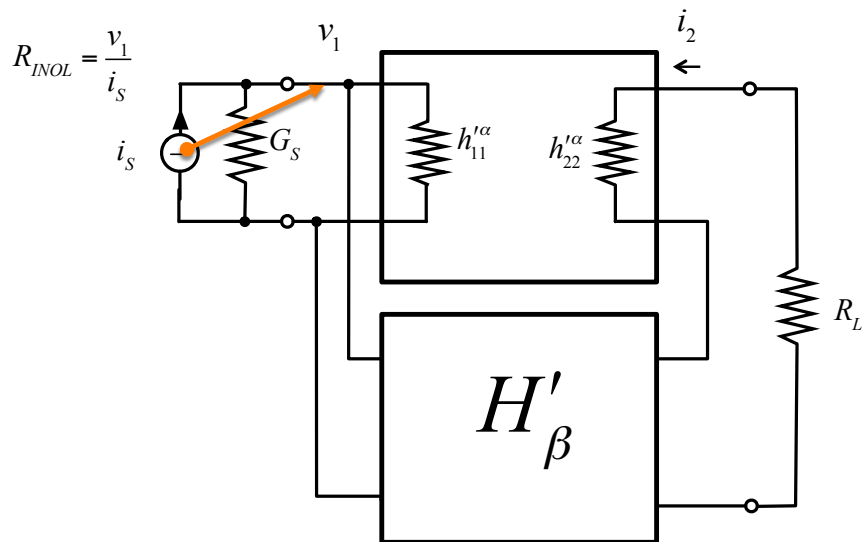
$$\frac{v_1}{i_s} = \frac{(h'_{22} + h'_{22} + R_L)}{\det H'_\gamma} = \frac{h'_{22} + h'_{22} + R_L}{(h'_{11} + h'_{11} + G_s)(h'_{22} + h'_{22} + R_L) - (h'_{12} + h'_{12})(h'_{21} + h'_{21})}$$

$$R_{IN} = \frac{h'_{22} + h'_{22} + R_L}{(h'_{11} + h'_{11} + G_s)(h'_{22} + h'_{22} + R_L) - (h'_{12} + h'_{12})h'_{21}} \cdot \frac{1}{1 - \frac{(h'_{12} + h'_{12})h'_{21}}{(h'_{11} + h'_{11} + G_s)(h'_{22} + h'_{22} + R_L) - (h'_{12} + h'_{12})h'_{21}}}$$

Resistenza anello aperto

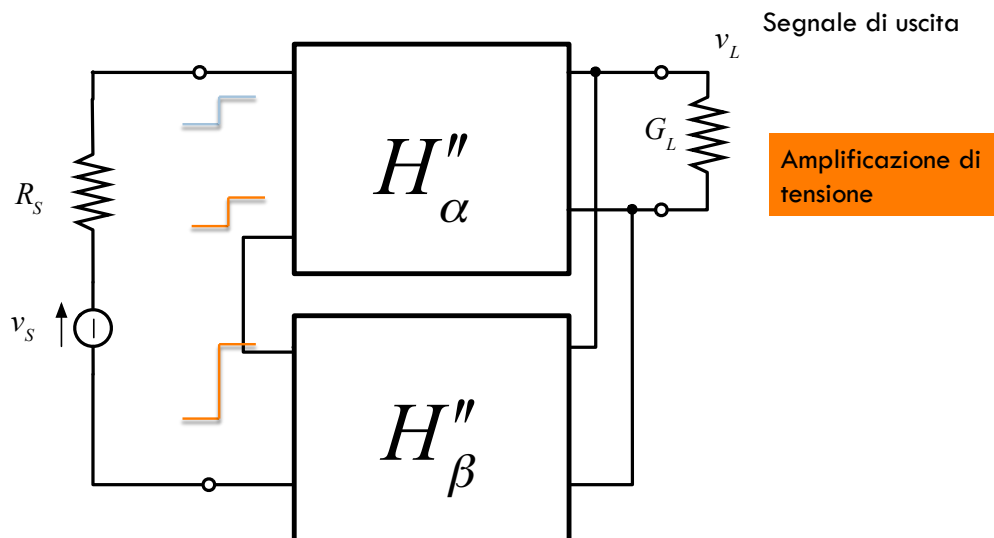
$$R_{IN} = R_{INOL} \frac{1}{1 - G_{loop}}$$

G_{an}ello aperto nel circuito



Retroazione serie parallelo

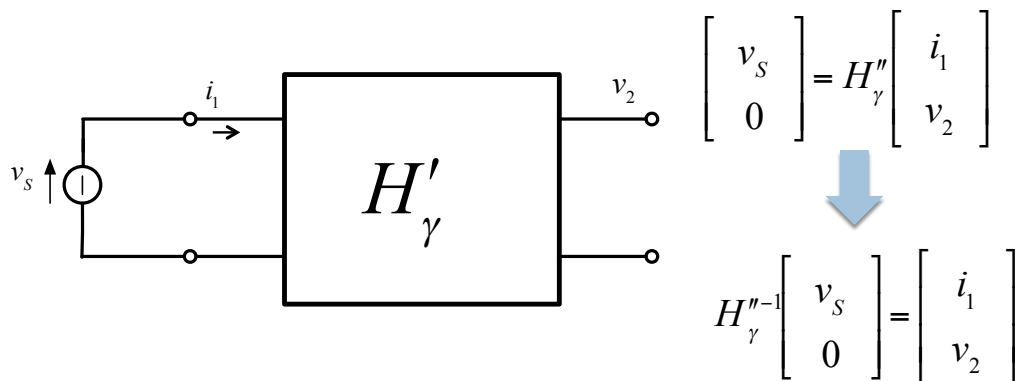
- Accesso ad un nodo del circuito retroazionato
l'uscita è su un ramo della retroazione



Retroazione parallelo parallelo

Calcolo del risultato

$$H''_{\gamma} = H''_{\alpha} + H''_{\beta} + \begin{bmatrix} R_S & 0 \\ 0 & G_L \end{bmatrix} = \begin{bmatrix} h''_{11}{}^{\alpha} + h''_{11}{}^{\beta} + R_S & h''_{12}{}^{\alpha} + h''_{12}{}^{\beta} \\ h''_{21}{}^{\alpha} + h''_{21}{}^{\beta} & h''_{22}{}^{\alpha} + h''_{22}{}^{\beta} + G_L \end{bmatrix}$$



Retroazione parallelo serie

Soluzione

$$\frac{1}{\det(H''_{\gamma})} \begin{bmatrix} h''_{22}{}^{\alpha} + h''_{22}{}^{\beta} + G_L & -(h''_{12}{}^{\alpha} + h''_{12}{}^{\beta}) \\ -(h''_{21}{}^{\alpha} + h''_{21}{}^{\beta}) & h''_{11}{}^{\alpha} + h''_{11}{}^{\beta} + R_S \end{bmatrix} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

Analisi del primo trasferimento

Termine dominante A_{VD}

Termine parassita A_{VP}

$$\frac{v_2}{v_s} = \frac{-(h''_{21}{}^{\alpha} + h''_{21}{}^{\beta})}{\det H''_{\gamma}} = \frac{-h''_{21}{}^{\alpha}}{\det H''_{\gamma}} + \frac{-h''_{21}{}^{\beta}}{\det H''_{\gamma}}$$

Trasferimento

Trasferimento

Termine dominante A_{VD}

Termine parassita A_{VP}

$$\frac{v_2}{v_S} = \frac{-(h_{21}^{\alpha\alpha} + h_{21}^{\alpha\beta})}{\det H_\gamma''} = \frac{-h_{21}^{\alpha\alpha}}{\det H_\gamma''} + \frac{-h_{21}^{\alpha\beta}}{\det H_\gamma''}$$

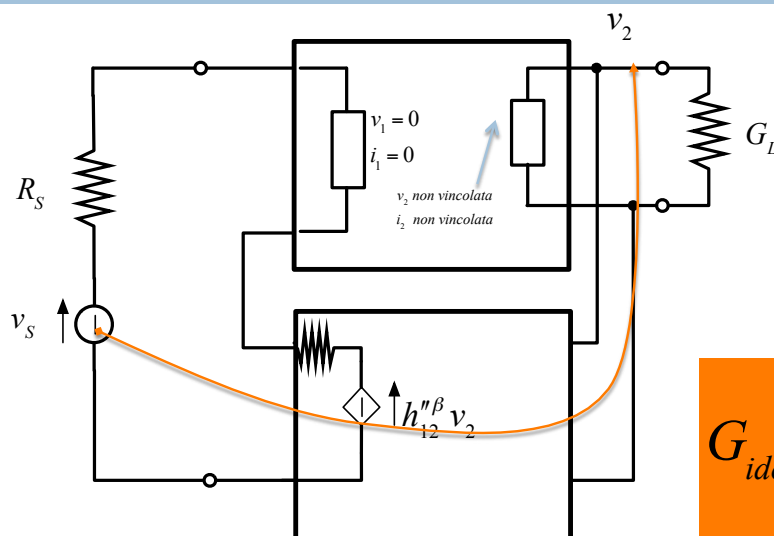
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$H_\alpha'' = \begin{bmatrix} R_{in} & \approx 0 \\ A_I & 1/R_{out} \end{bmatrix}$$

$$G_{ideale} = \lim_{A_I \rightarrow \infty} A_{VD} = \lim_{A_I \rightarrow \infty} \frac{-A_I}{(h_{11}^{\alpha\alpha} + h_{11}^{\alpha\beta} + R_S)(h_{22}^{\alpha\alpha} + h_{22}^{\alpha\beta} + G_L) - (h_{12}^{\alpha\alpha} + h_{12}^{\alpha\beta})(A_I + h_{21}^{\alpha\beta})} = \frac{1}{(h_{12}^{\alpha\alpha} + h_{12}^{\alpha\beta})} \approx \frac{1}{h_{12}^{\alpha\beta}}$$

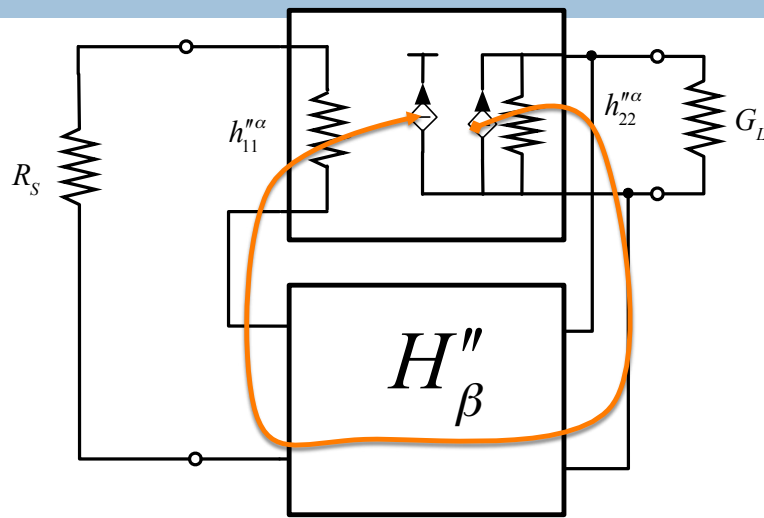
$$A_{VD} = G_{ideale} \frac{1}{1 + \frac{h_{21}^{\alpha\beta}(h_{12}^{\alpha\alpha} + h_{12}^{\alpha\beta}) - (h_{11}^{\alpha\alpha} + h_{11}^{\alpha\beta} + R_S)(h_{22}^{\alpha\alpha} + h_{22}^{\alpha\beta} + G_L)}{(h_{12}^{\alpha\alpha} + h_{12}^{\alpha\beta})h_{21}^{\alpha\alpha}}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



$$G_{ideale} = \frac{v_2}{v_S}$$

G_{loop} nel circuito



$$G_{loop} = \frac{-\left(h''_{12} + h''_{12}\right)h''_{21}}{h''_{21}\left(h''_{12} + h''_{12}\right) - \left(h''_{11} + h''_{11} + R_S\right)\left(h''_{22} + h''_{22} + G_L\right)}$$

Analisi del secondo termine

□ Analisi del secondo termine

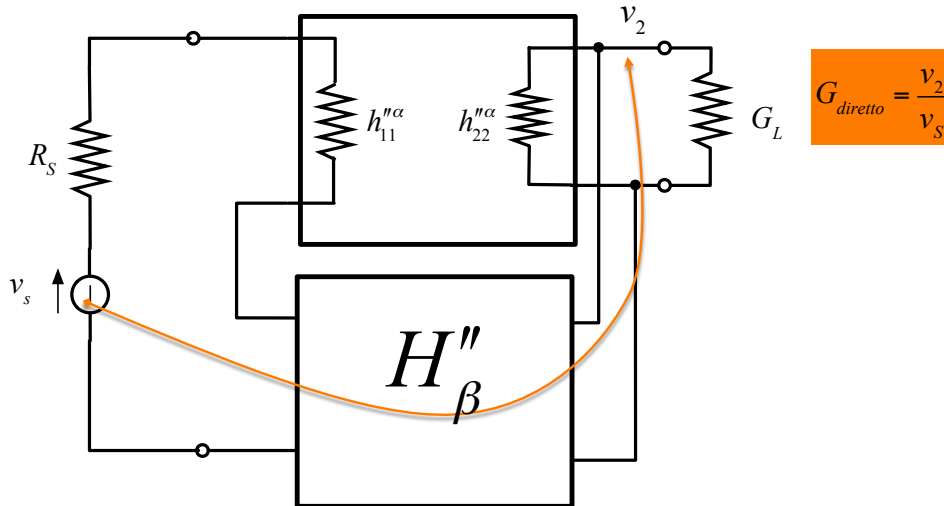
Guadagno diretto

$$A_{VP} = \frac{h''_{21}}{h''_{21}\left(h''_{12} + h''_{12}\right) - \left(h''_{11} + h''_{11} + R_S\right)\left(h''_{22} + h''_{22} + G_L\right)} \cdot \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{h''_{21}}{h''_{21}\left(h''_{12} + h''_{12}\right) - \left(h''_{11} + h''_{11} + R_S\right)\left(h''_{22} + h''_{22} + G_L\right)}$$

$$A_{VP} = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(H''_\gamma)} \begin{bmatrix} h''_{22} + h''_{22} + G_L & -(h''_{12} + h''_{12}) \\ -(h''_{21} + h''_{21}) & h''_{11} + h''_{11} + R_s \end{bmatrix} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

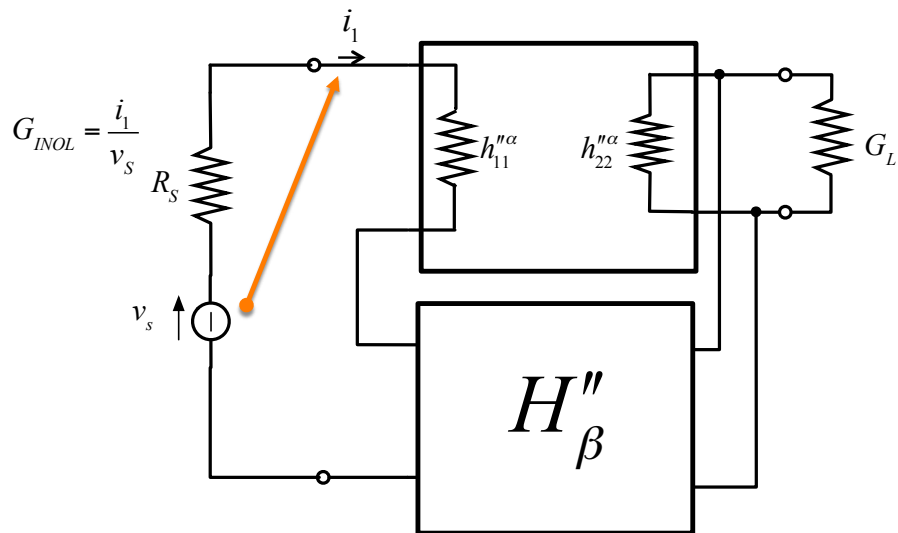
$$\frac{i_1}{v_s} = \frac{(h''_{22} + h''_{22} + G_L)}{\det H''_\gamma} = \frac{h''_{22} + h''_{22} + G_L}{(h''_{11} + h''_{11} + R_s)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})(h''_{21} + h''_{21})}$$

$$G_{IN} = \frac{h''_{22} + h''_{22} + G_L}{(h''_{11} + h''_{11} + R_s)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})h''_{21}} \cdot \frac{1}{1 - \frac{(h''_{12} + h''_{12})h''_{21}}{(h''_{11} + h''_{11} + R_s)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})h''_{21}}}$$

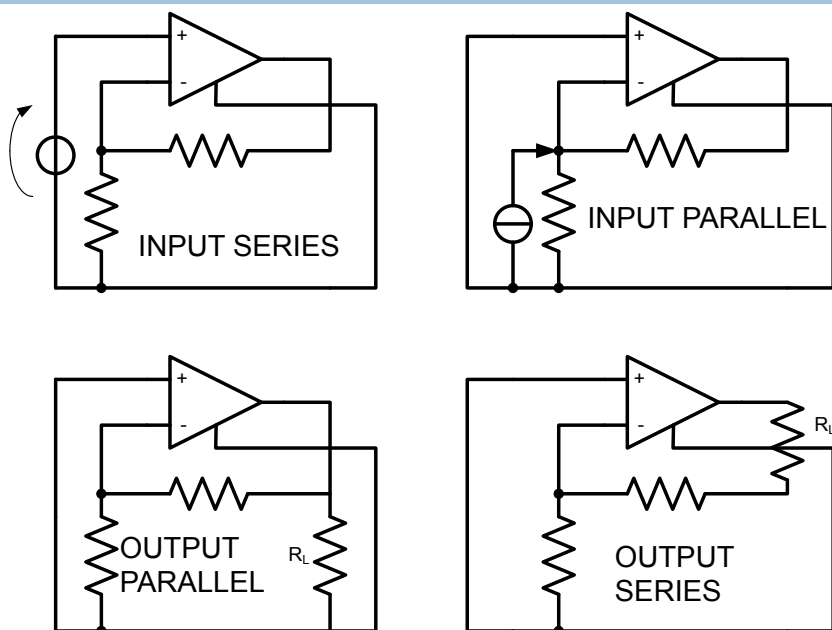
Conduttanza anello aperto

$$G_{IN} = G_{INOL} \frac{1}{1 - G_{loop}}$$

G_{IN} ad anello aperto nel circuito



Generazione delle varie combinazioni sullo stesso circuito



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CHARCATERISTICS OF OPERATIONAL AMPLIFIERS

Assistant professor Stefano Saggini
Nome della conferenza o seminario

Different types of Operational amplifiers

- ☐ Voltage mode operational amplifier
- ☐ Current feedback
- ☐ Norton operational amplifier

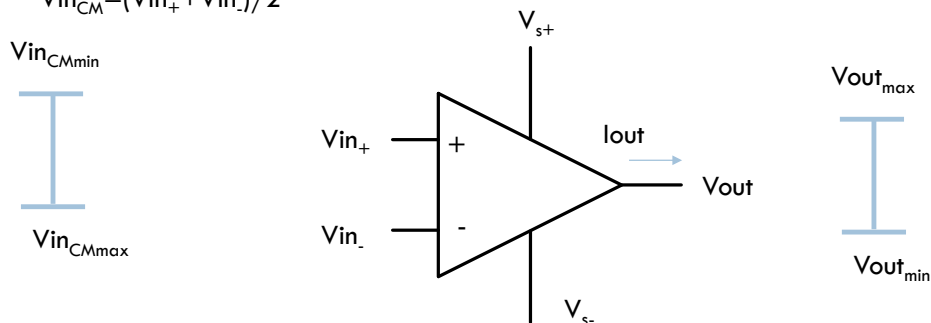
OPAMP general Characteristics

- Polarization and/or large signal
 - Input common mode range $ICMR$
 - Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$
 - Maximum output current
 - Input bias current
 - Slew rate SR
- AC characteristics
 - Input impedance
 - DC Gain $A_v(0)$
 - Output impedance
 - Gain band product
 - $CMRR$
 - $PSRR$
 - Input equivalent noise
- Mismatch non idealities
 - Input offset V_{offset}
 - Input current mismatch

Polarization and/or large signal

- Input common mode range $ICMR$
- Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$

$$V_{in_{CM}} = (V_{in_{+}} + V_{in_{-}}) / 2$$



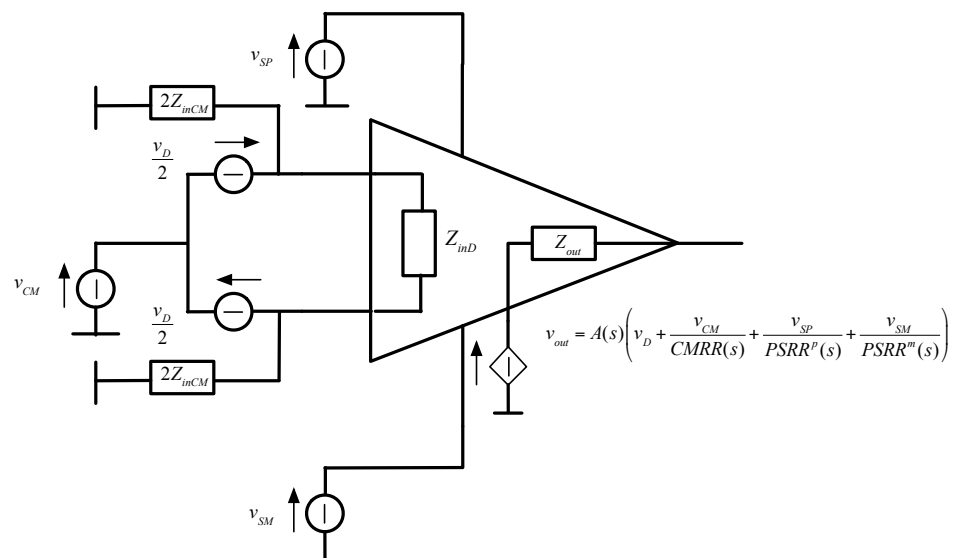
Se $V_{in_{CM}}$ è nel range $(V_{in_{CMmin}}, V_{in_{CMmax}})$, se V_{out} è nel range $(V_{out_{min}}, V_{out_{max}})$ e se la corrente di uscita l_{out} è nel range $(l_{out_{min}}, l_{out_{max}})$ dove $l_{out_{min}}$ è la massima corrente entrante (Massimo sinking) e $l_{out_{max}}$ è la massima corrente uscente (Massimo sourcing).

L'amplificatore operazionale è polarizzato correttamente e può essere studiato con il suo modello di piccolo segnale.

AC linear model Small Signal

- Input impedance differential
- Output impedance
- Gain transfer function (GBWP)
- CMRR
- PSRR

Scheme

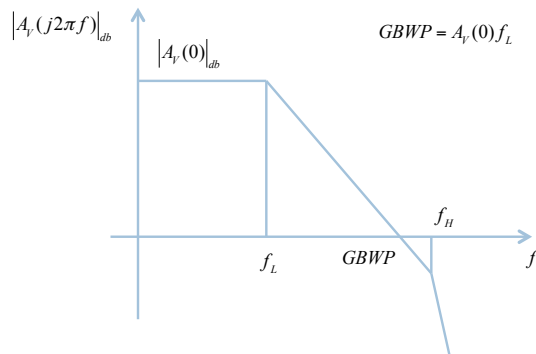


Gain transfer function

Transfer function

$$\frac{v_{OUT}}{v_D} = A_v(s)$$

$$A_v(s) = \frac{A_v(0)}{(1 + s\tau_L)(1 + s\tau_H)}$$



Example CMRR

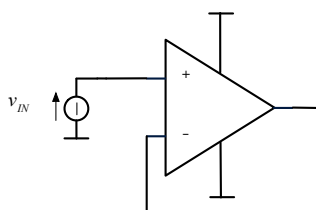
Considering CMRR in the transfer function

$$\frac{v_{OUT}}{v_{IN}} = \frac{G_{ideale}}{1 - G_{loop}^{-1}}$$

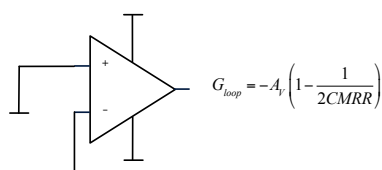
$$v_{in} - v_{out} + \frac{v_{in} + v_{out}}{2CMRR} = 0 \quad \Rightarrow \quad \frac{v_{out}}{v_{in}} = \frac{1 + \frac{1}{2CMRR}}{1 - \frac{1}{2CMRR}}$$

$$\frac{v_{OUT}}{v_{IN}} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} = \frac{A_v \left(1 + \frac{1}{2CMRR} \right)}{1 + A_v \left(1 - \frac{1}{2CMRR} \right)}$$

$$\frac{v_{out} - v_{in}}{v_{in}} = \frac{1 + \frac{1}{2CMRR}}{1 - \frac{1}{2CMRR}} - 1 = \frac{\frac{1}{CMRR}}{1 - \frac{1}{2CMRR}} \approx \frac{1}{CMRR}$$

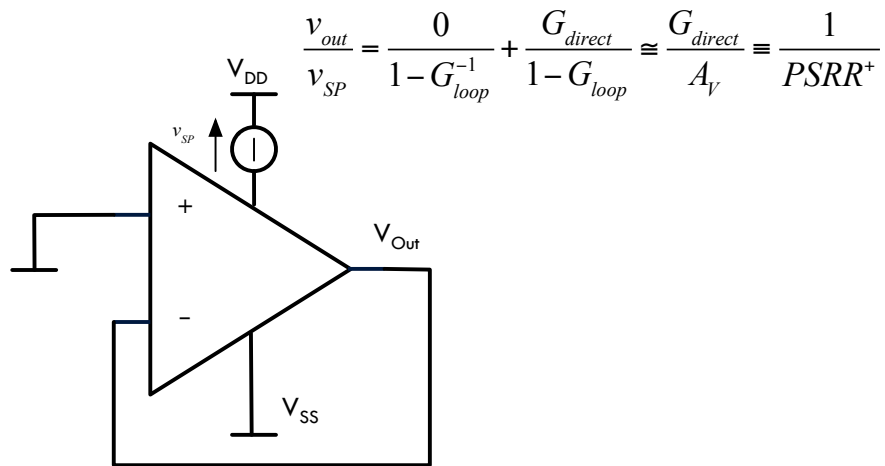


Gloop



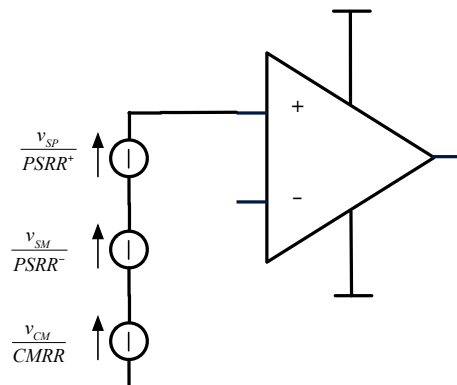
Example PSRR

- Signal noise on vdd



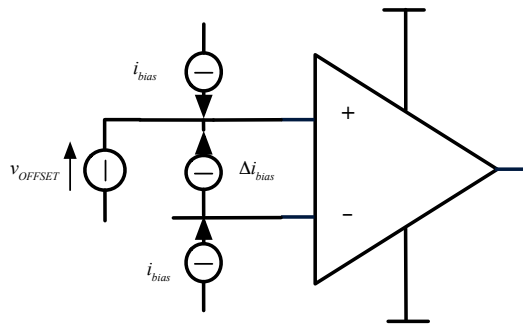
Equivalent circuit

- Equivalent circuit



Effetto di offset e bias

- Circuito equivalente



Limitazioni di grande segnale

- Limitazione dovuta alla SR considerando un segnale sinusoidale di uscita di ampiezza A e frequenza ω

$$A\omega < SR$$

- Limitazione dovuta alla I_{out} sulla massima derivata di uscita

$$A\omega < \frac{I_{out}}{C_L}$$

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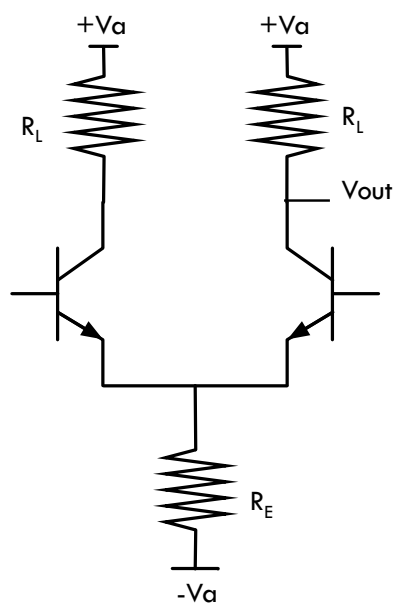
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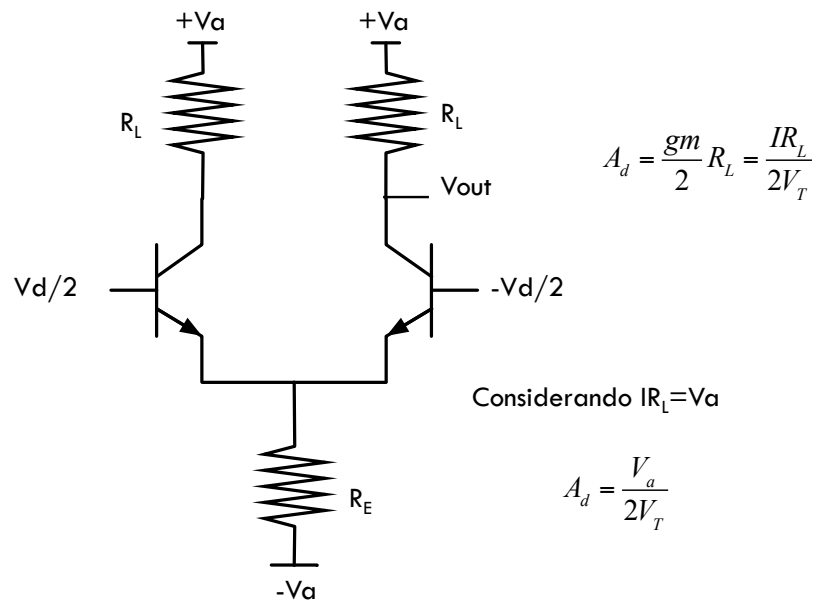
BASIC ANALOG STAGES

Associate professor Stefano Saggini
Lezione di circuiti e sistemi elettronici

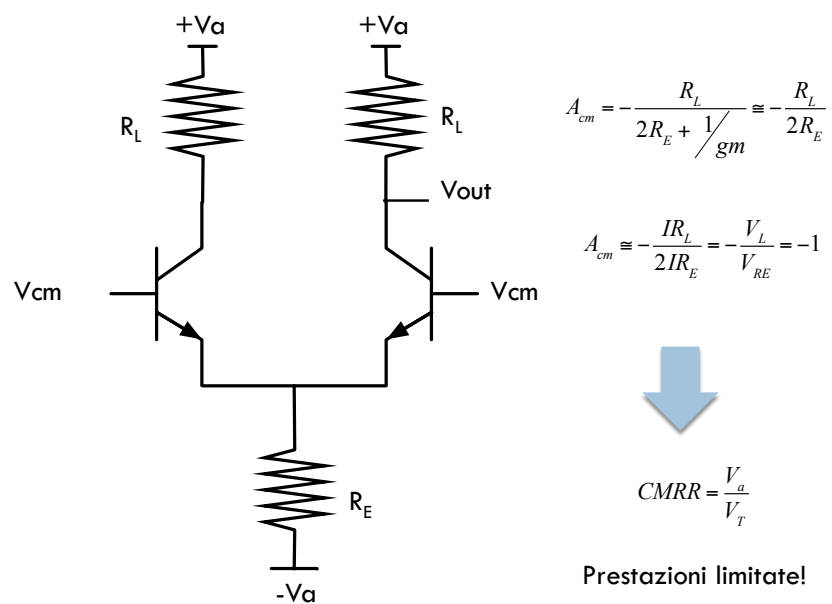
Stadi differenziali



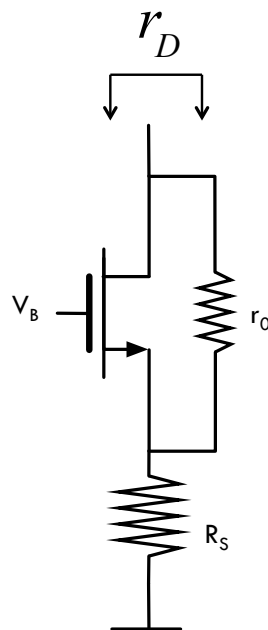
Stadi differenziali



Stadi differenziali



Generatori di corrente



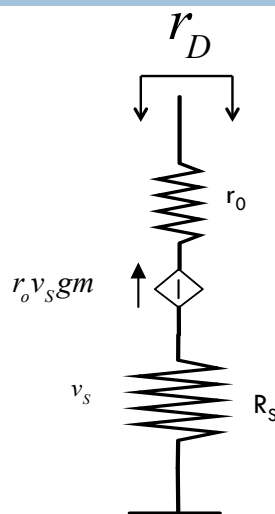
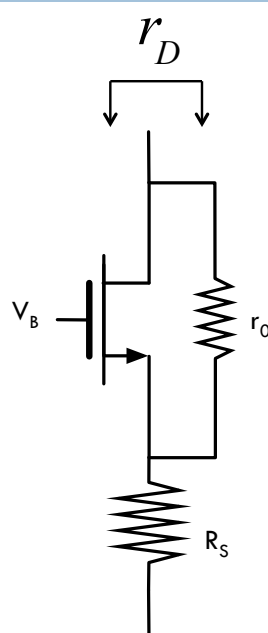
Positive loop approach

$$r_D = \frac{r_o + 1/gm // R_S}{1 - Gloop} = \frac{r_o + 1/gm // R_S}{1 - \frac{R_S}{1/gm + R_S}} = r_o(R_S gm + 1) + R_S$$

Open loop approach (forzo in tensione con VD)

$$i_D = \frac{v_D}{r_o + 1/gm // R_S} \left(1 - \frac{R_S}{1/gm + R_S} \right)$$

Generatori di corrente

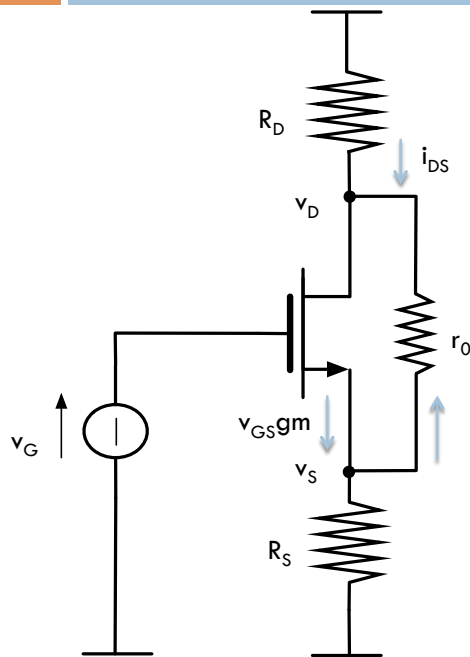


Approccio retroazione serie-serie

$$r_D = R_{OL}(1 - Gloop) = (R_S + r_o) \left(1 + \frac{gm r_o R_S}{R_S + r_o} \right)$$

$$r_D = R_S + r_o (1 + gm R_S)$$

Guadagno stadi elementari con r_o

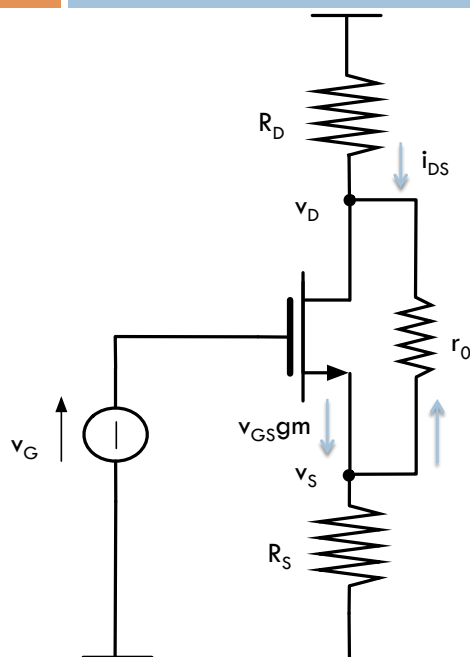


$$i_{DS} = v_{GS} gm + \frac{v_{DS}}{r_o}$$

$$i_{DS} = v_{GS} gm - i_{DS} \frac{R_S + R_D}{r_o}$$

$$i_{DS} = \frac{v_{GS}}{\frac{1}{gm} + \frac{R_S + R_D}{r_o gm}} = \frac{v_{GS}}{gm \left(1 + \frac{R_S + R_D}{r_o} \right)}$$

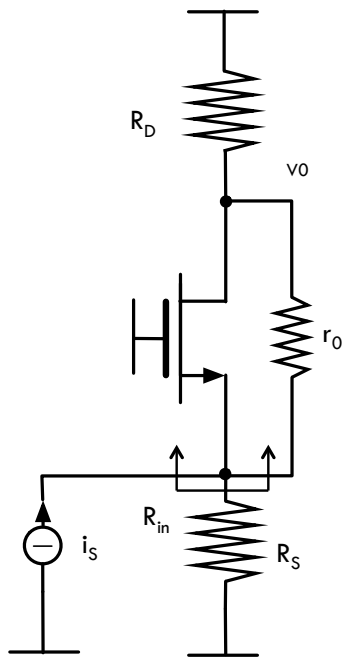
Guadagno stadi elementari con r_o



$$\frac{v_D}{v_G} = \frac{-R_D}{\frac{1}{gm} \left(1 + \frac{R_S + R_D}{r_o} \right) + R_S}$$

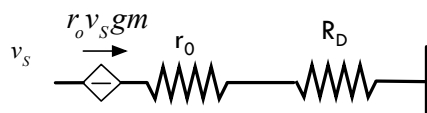
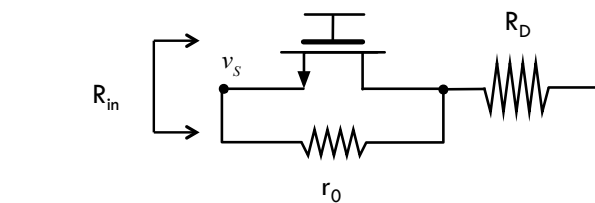
$$\frac{v_S}{v_G} = \frac{R_S}{\frac{1}{gm} \left(1 + \frac{R_S + R_D}{r_o} \right) + R_S}$$

Guadagno stadi elementari con r_o



$$v_o = i_s \frac{R_S}{R_S + R_{in}} R_D$$

Guadagno stadi elementari con r_o

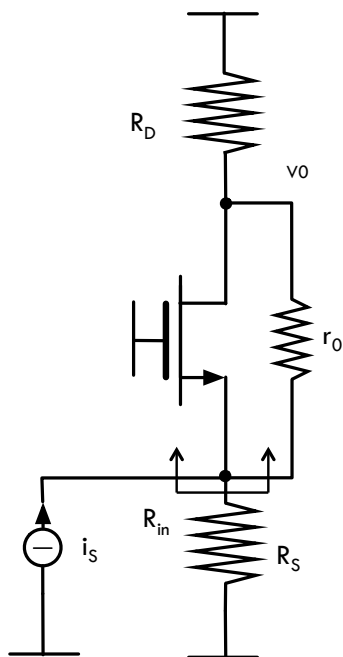


Retroazione negativa (analogamente a prima può essere interpretata utilizzando il modello con i generatori di corrente come retroazione positiva minore di 1)

$$R_{in} = \frac{r_o + R_D}{1 - \text{Gloop}(0)} = \frac{r_o + R_D}{1 + gmr_o}$$

Ovviamente tende a $1/gm$ se r_o è molto alta e maggiore di R_D .

Guadagno stadi elementari con r_o



$$v_o = i_s \frac{R_S}{R_S + R_{in}} R_D$$

$$v_o = i_s \frac{R_S}{R_S + \frac{r_o + R_D}{1 + g_m r_o}} R_D$$

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CURRENT MIRRORS

Current Mirrors

Small signal consideration

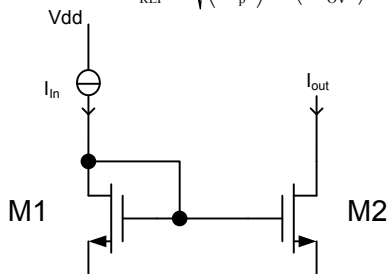
$$R_{in} = 1/gm_1$$

$$R_{out} = r_{o2}$$

$$V_{min} = V_{ov2}$$

Considering equal the two drain voltage

$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$

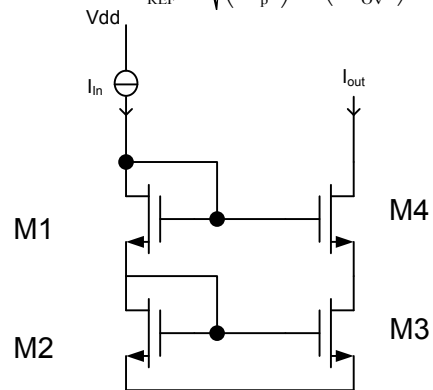


$$R_{in} = gm_1 gm_2 / (gm_1 + gm_2)$$

$$R_{out} = r_{o4} (1 + gm_4 r_{o3}) + r_{o3}$$

$$V_{min} = V_{gs3} + V_{ov4}$$

$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Mirror

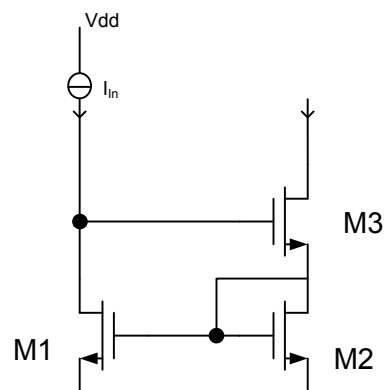
$$R_{in} \cong gm_2 gm_3 / (gm_3 + gm_2)$$

$$R_{out} \cong r_{o3} (1 + r_{o1} gm_3) \text{ with } (gm_1 \text{ equal to } gm_2)$$

$$V_{min} = V_{gs2} + V_{ov3}$$

Considering equal the two drain voltage

$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Mirror

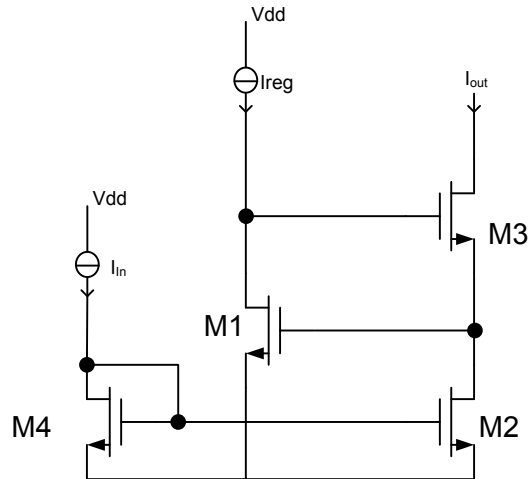
$$R_{in} = 1/gm_4$$

$$R_{out} \approx gm_3 r_{o3} r_{o2} gm_1 r_{o1}$$

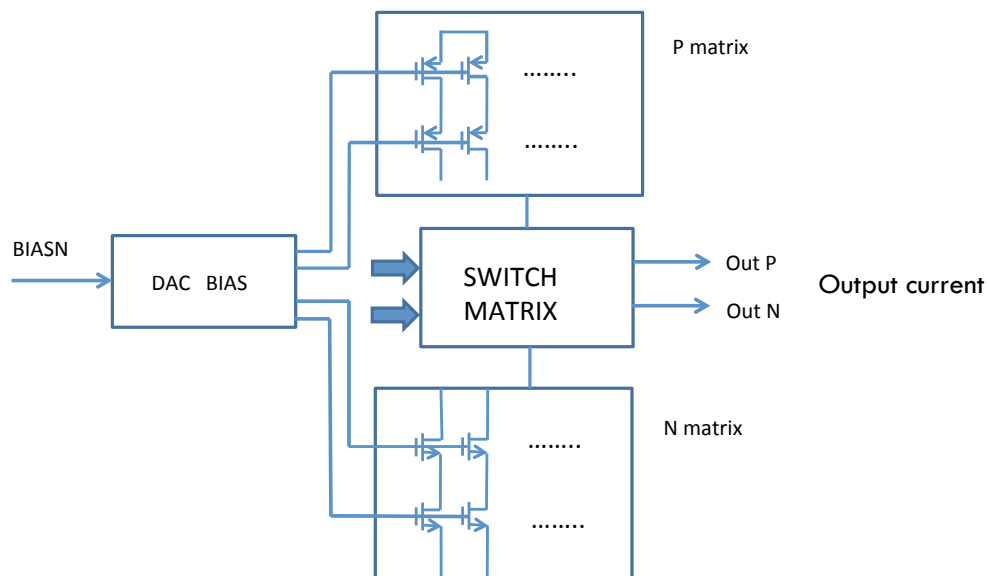
$$V_{min} = V_{ov3} + V_{gs1}$$

Considering equal the two drain voltage

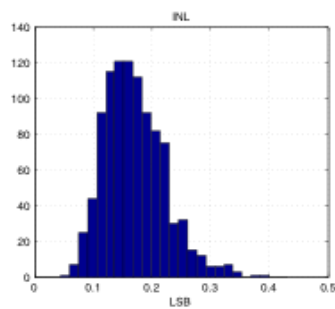
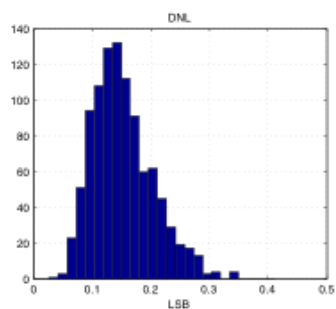
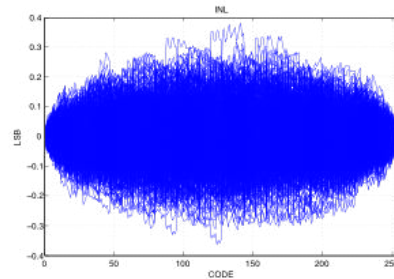
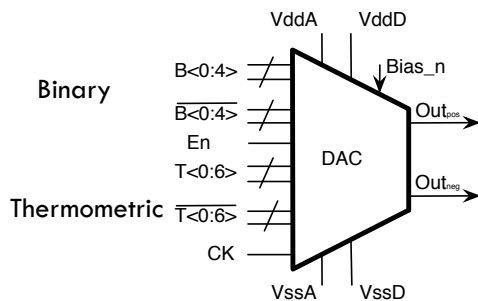
$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Steering DAC



Current Steering DAC



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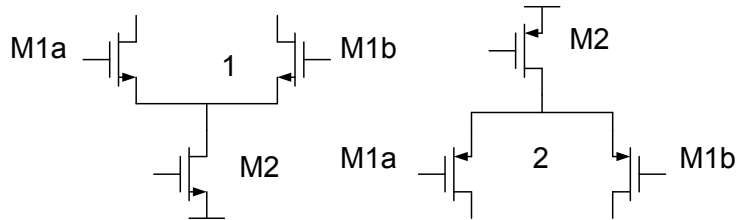
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SECOND STAGE

Assistant professor Stefano Saggini
Nome della conferenza o seminario

Differential stage



Input Dynamic

ICMR

1. $V_{incm} (min) = V_{ov1} + V_{ov2} + V_{Tn}$ (for n MOS) V_{ss} for (for p MOS)
2. $V_{incm} (max) = V_{dd} - V_{ov1} - V_{ov2} - |V_{Tp}|$ (for p MOS) V_{dd} for (for n MOS)

Trasconduttance :

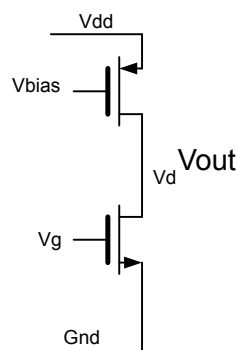
$$gm = \sqrt{K_1 I_{D2}}$$

Precision

$$\sigma_{Vos} = \sqrt{\sigma_{Vt1,2}^2 + \left(\frac{V_{OV1,2}}{2}\right)^2 \left[\left(\frac{\sigma_{Kp}}{k_p}\right)_{a,b}^2 + \left(\frac{\sigma_I}{I}\right)^2 \right]}$$

Specchio

Inverter current source



Output signal range

$$V_{out_{max}} = V_{dd} - V_{ovp}$$

$$V_{out_{min}} = V_{ovn}$$

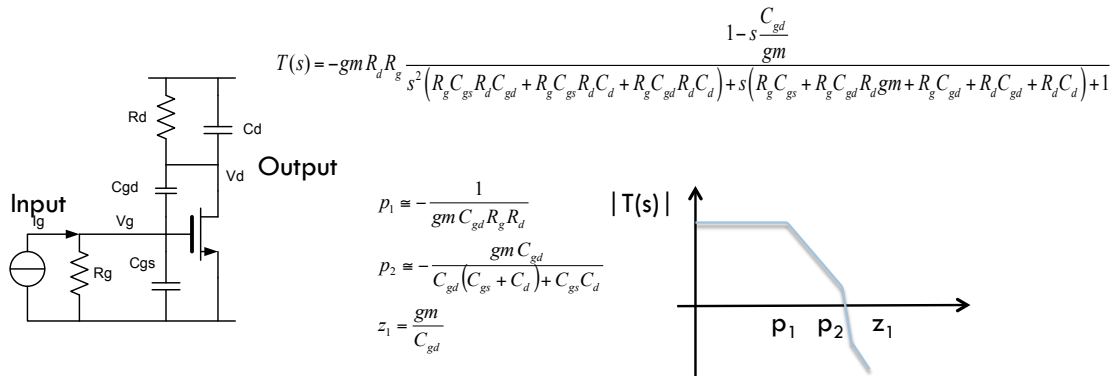
Input dynamics

$$V_{in_{min}} \cong V_{in_{max}} \cong V_{ovn} + VT$$

To increase the dynamics

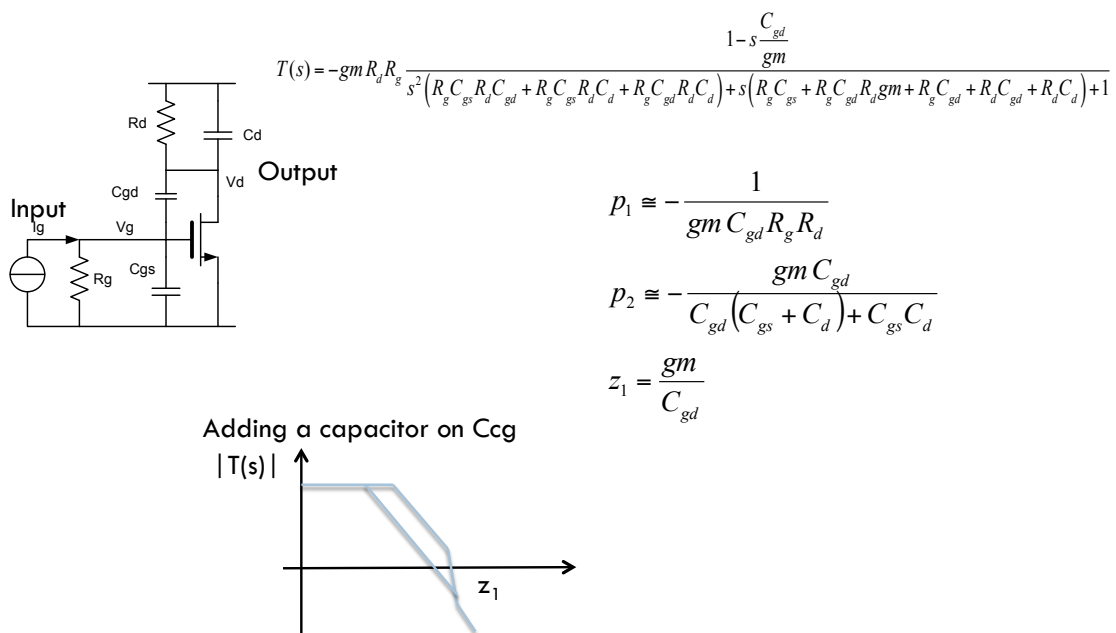
- Increase W/L of nch and pch (decrease V_{ov})
- Reduce the current

Inverter current source Tf



- The dominant pole p_1 can be calculated by using miller theorem
- To increase the performance
 - ▢ Increase W/L of nch increase g_m and C_{gd}
 - ▢ Increase the current (increase of g_m)

Pole splitting



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DESIGN OF OPAMPS

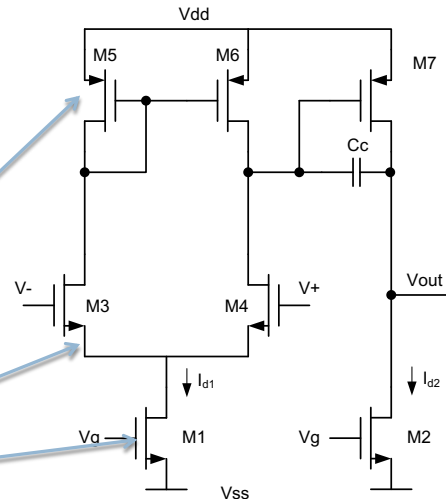
Assistant professor Stefano Saggini
Nome della conferenza o seminario

OPAMP general Specifications

- DC Gain $A_v(0)$
 - ▣ *Also the (depend on the application) CMRR*
- Gain band product
- Input offset V_{offset}
- Input common mode range $ICMR$
- Output capacitance C_L
- Slew rate SR
- Output voltage swing $V_{out_{max}}$ $V_{out_{min}}$
- Consumption P_{diss}
- Input equivalent noise

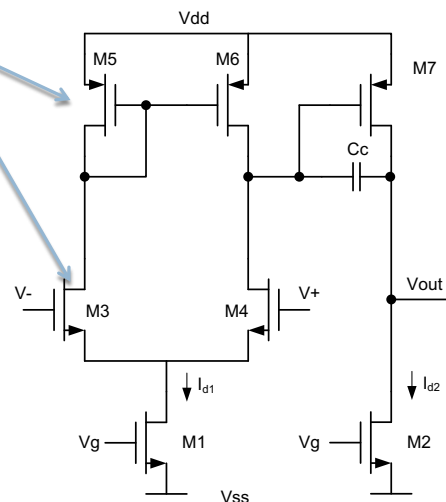
Two stage OPAMP design Steps

- Depending on the ICMR spec choice of the n-ch differential stage or the p-ch
- Determination of the V_{ov} of M5 and M6 for the Max ICMR
- Determination of the V_{ov} of M1,2,3,4 Min ICMR



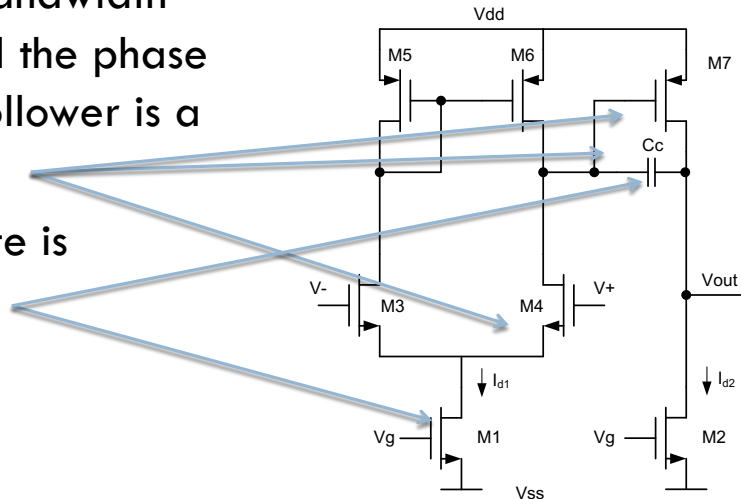
Two stage OPAMP design Steps

- The Input offset
determine the
dimension of the stage



Two stage OPAMP design Steps

- The Gain bandwidth product and the phase margin in follower is a function of
- The slew rate is function of



Project of two stage OpAmp

- Determination of the capacitor C_c

$$GB \cong \frac{gm_{3,4}}{C_c}$$

$$p_1 \cong \frac{-1}{\left(\frac{r_{o7} r_{o2}}{r_{o7} + r_{o2}}\right) gm_7 \left(\frac{r_{o6} r_{o4}}{r_{o6} + r_{o4}}\right) C_c}$$

$$p_2 \cong \frac{-gm_7}{C_L}$$

$$z_1 = \frac{gm_7}{C_c}$$

$$60 < 180 - \tan^{-1}\left(\frac{GB}{p_1}\right) - \tan^{-1}\left(\frac{GB}{p_2}\right) - \tan^{-1}\left(\frac{GB}{z_1}\right)$$

Considering the ratio 0.1

$$gm_7 > 10 gm_{3,4}$$

$$|p_2| > 2.2 GB$$

$$\frac{gm_7}{C_L} > 2.2 \frac{gm_{3,4}}{C_c}$$

$$C_c > 0.22 C_L$$

Project of two stage OpAmp

- Determination of $gm_{3,4}$

$$gm_{3,4} \cong GB C_c$$

- From slew rate can be determined the I_{d1}

$$I_{d1} = SRCc$$

- From the two information can be calculated the $V_{ov_{3,4}}$

$$V_{ov_{3,4}} = \frac{2I_D}{gm_{3,4}} = \frac{I_{d1}}{gm_{3,4}}$$

Project of two stage OpAmp

- Determination of the ratio of 3,4

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2\left(\frac{I_{d1}}{2}\right)}{K'_n[V_{ov_{3,4}}]^2}$$

- As a function of the ICMR max the $(W/L)_{5,6}$ is determined

$$\left(\frac{W}{L}\right)_{5,6} = \frac{2\left(\frac{I_{d1}}{2}\right)}{K'_p[V_{DD} - V_{in}(\max) - |V_{T5}|(\max) + V_{T3}(\min)]^2}$$

Project of two stage OpAmp

- As a function of the offset we will determine the values of Area

Minimum area required

$$\sigma_{V_{os}} = \sqrt{\sigma_{V_{tp3,4}}^2 + \left(\frac{V_{OV3,4}}{2}\right)^2 \left[\left(\frac{\sigma_{K_p}}{k_p}\right)_{3,4}^2 + \left(\frac{\sigma_{I_D}}{I_D}\right)_{5,6}^2 \right]}$$

$$\left(\frac{\sigma_{I_D}}{I_D}\right)_{5,6} = \sqrt{\left(\frac{\sigma_{K_p}}{k_p}\right)_{5,6}^2 + \left(\frac{2\sigma_{V_{Tp5,6}}}{V_{OV5,6}}\right)^2}$$

$W_{3,4,5,6}$

$L_{3,4,5,6}$

Project of two stage OpAmp

- As a function of the ICMR min the $(W/L)_1$ is determined

$$V_{in\ min} = V_{ov1} + V_{ov3,4} + V_{T\ max}$$

$$\left(\frac{W}{L}\right)_1 = \frac{2(I_{d1})}{K'_n [V_{ov1}]^2}$$

Project of two stage OpAmp

- Considering the transconductance relation

$$gm_7 > 10 gm_{3,4}$$

- Considering

$$V_{ov_{5,6}} = V_{ov_7}$$

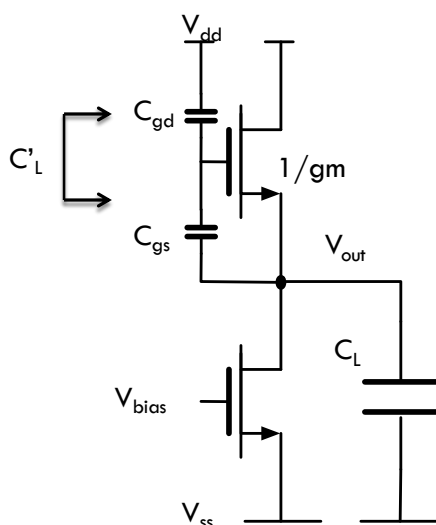
$$\left(\frac{W}{L}\right)_7 > \frac{10 gm_{3,4}}{K_P V_{ov_{5,6}}}$$

$$Id_7 = Id_2 = \left(\frac{W}{L}\right)_7 / \left(\frac{W}{L}\right)_{5,6} \frac{Id_1}{2}$$

$$\left(\frac{W}{L}\right)_2 = \left(\frac{W}{L}\right)_1 \frac{Id_2}{Id_1}$$

Output stage

- Obviously a buffer output stage reduce C_c



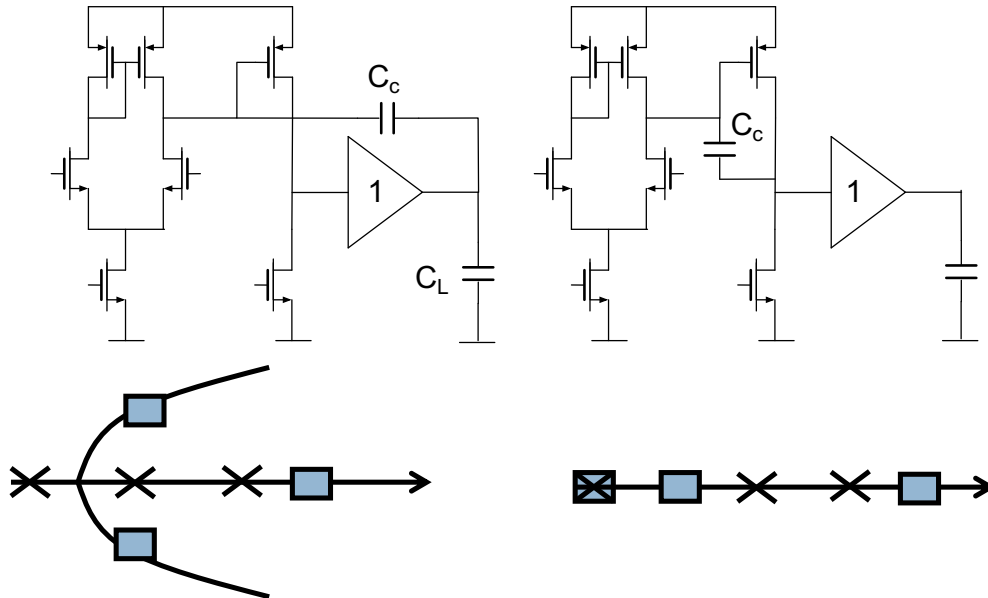
$$C'_L < C_L$$

Condition to be imposed

$$|p_3| \cong \frac{1}{R_{out} C_L} = \frac{gm_{out}}{C_L} > 10 GB$$

Output stage

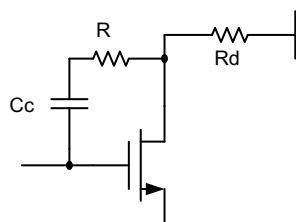
□ Output stage



Inserting a zero on the loop transfer function

□ Inserting the zero in the transfer function

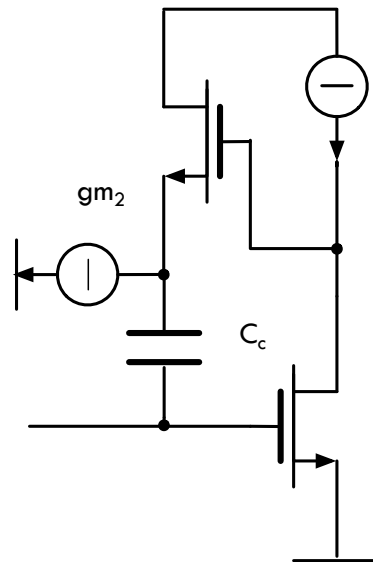
Second Stage of the OpAmp



$$z_1 = \frac{1}{C_c (1/gm - R)}$$

Inserting a zero on the loop transfer function

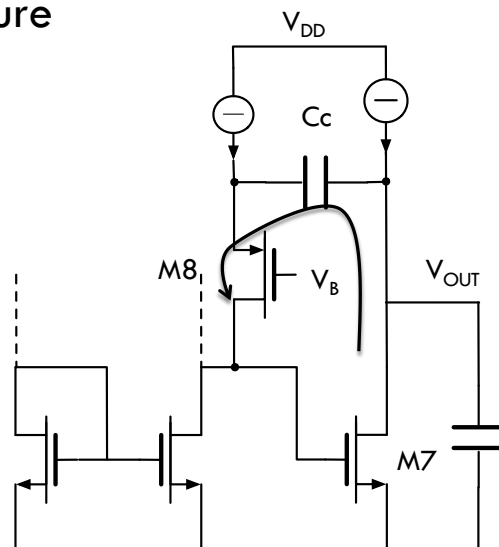
- Controlling the right half-plane zero



$$z_1 = -\frac{1}{C_c (1/gm_2)}$$

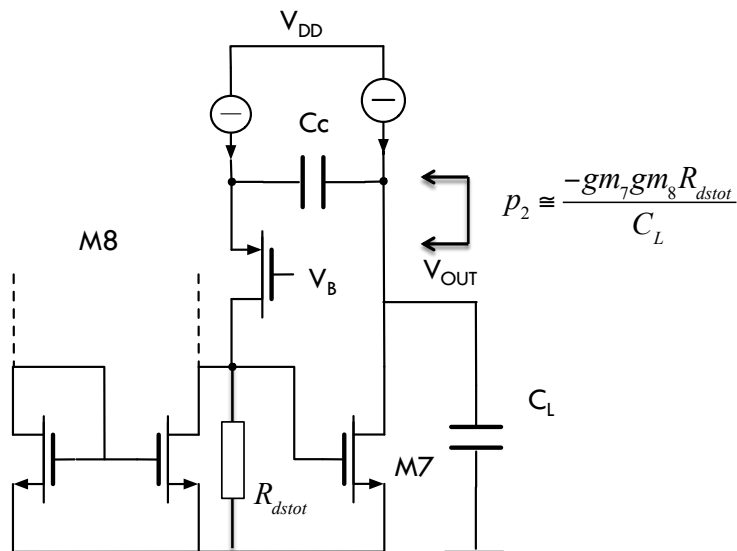
Compensation that increase p_2

- The feedback gain reduce the output impedance of the structure



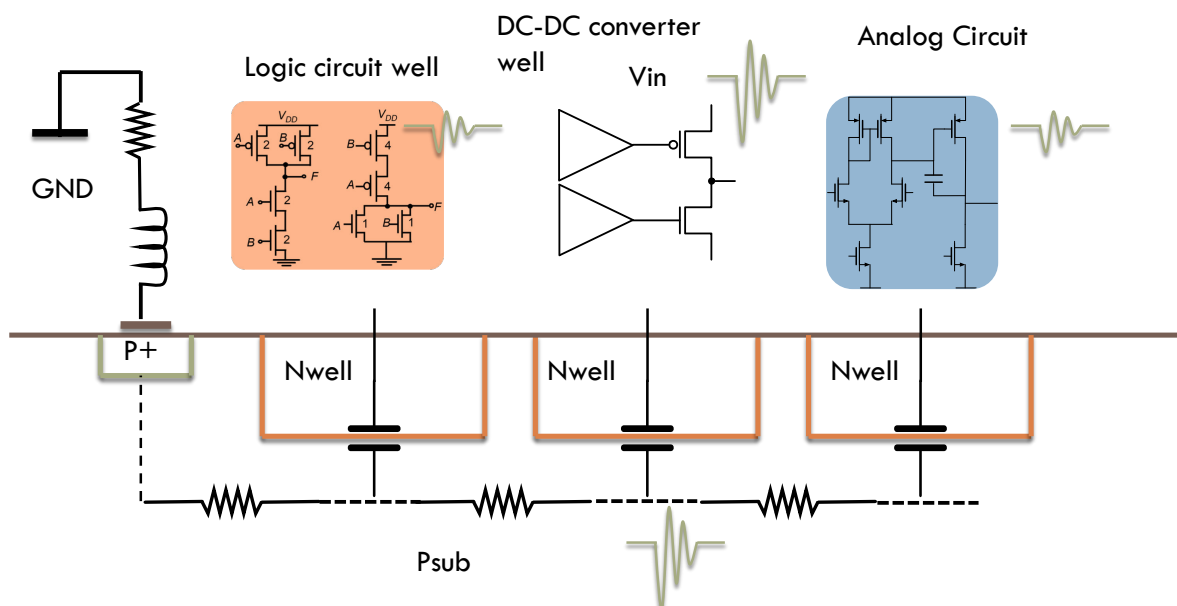
Compensation that increase p_2

- At high frequency



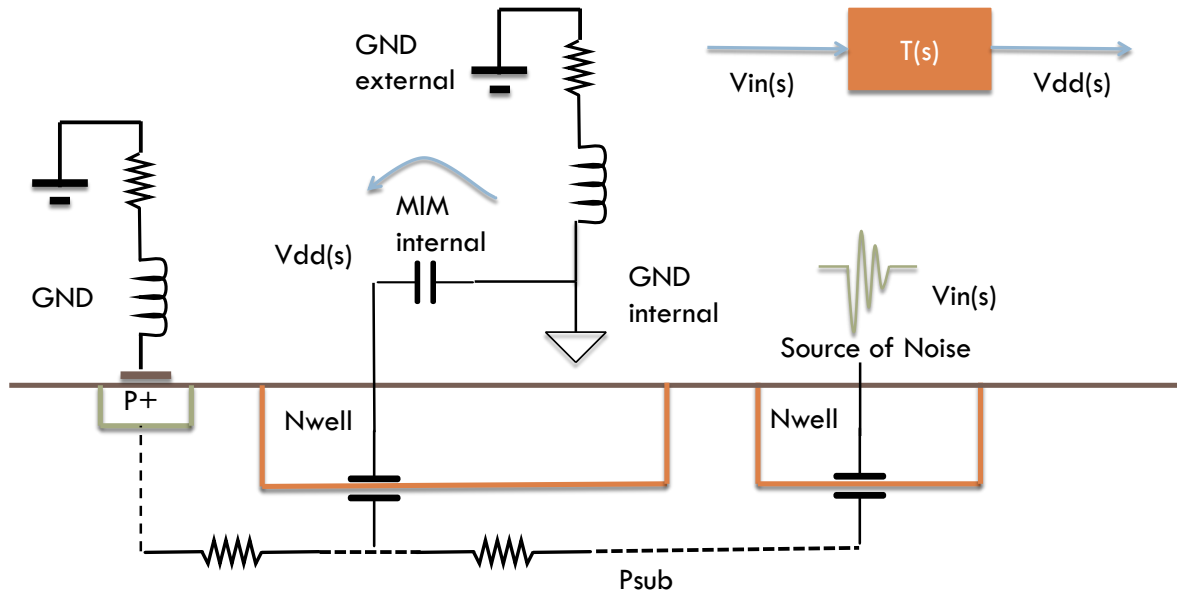
PSRR: Substrate noise coupling

- Vdd and Gnd bounce



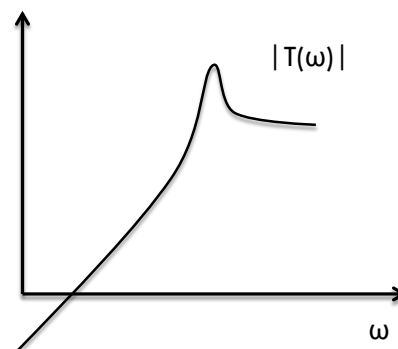
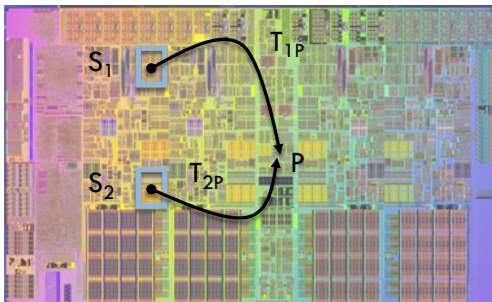
PSRR: Substrate noise coupling Analysis

- For each point can be defined a $T(s)$



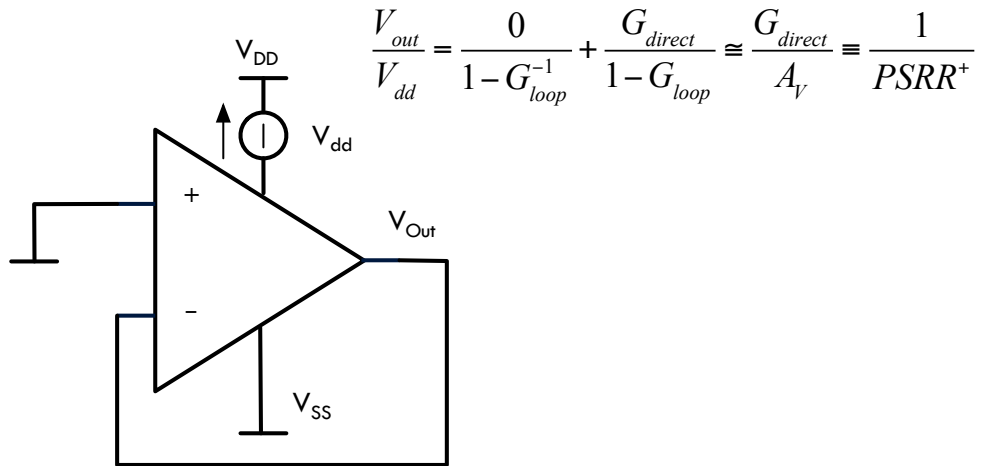
PSRR: Substrate noise coupling Analysis

- For each sensitive point and each noise source can be defined a $T(s)$



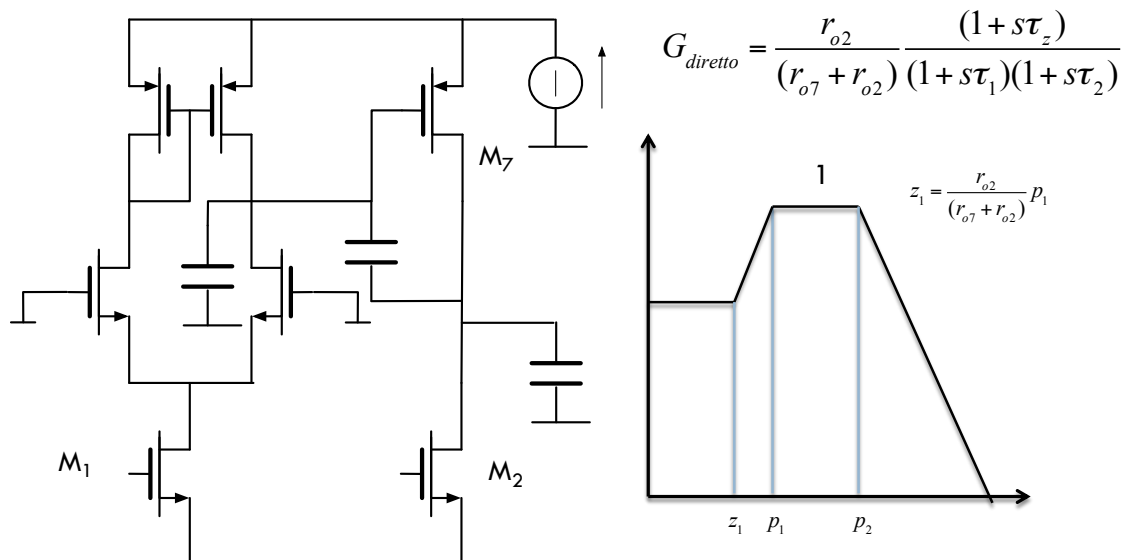
PSRR

Definition



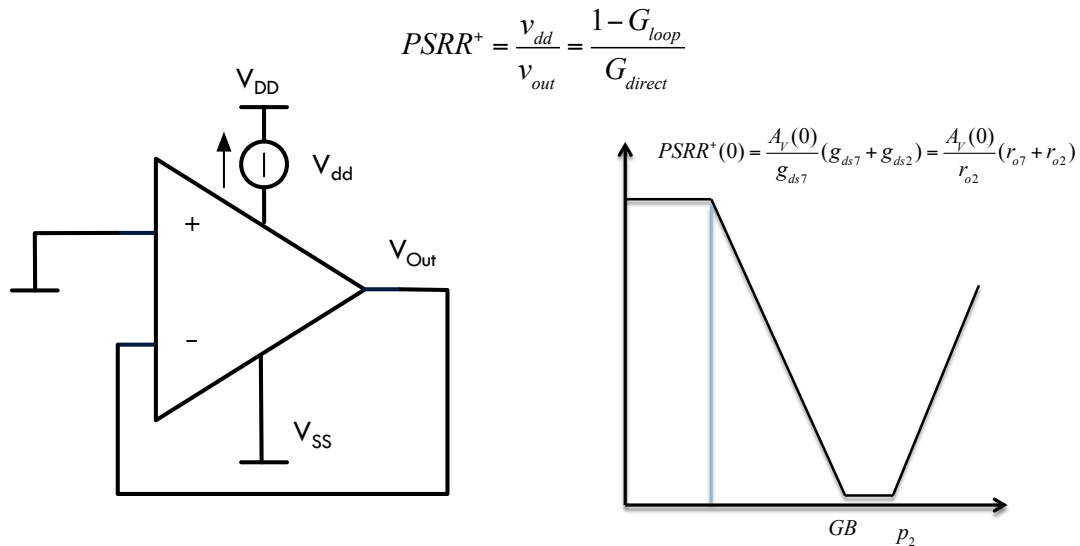
PSRR

Calculation



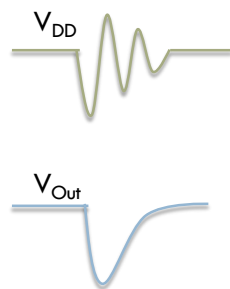
PSRR close loop

□ Calculation close loop



PSRR

- Non Linear effect of the OpAmp as Asymmetrical slew rate can rectify the noise
- This effect are studied by AC simulations

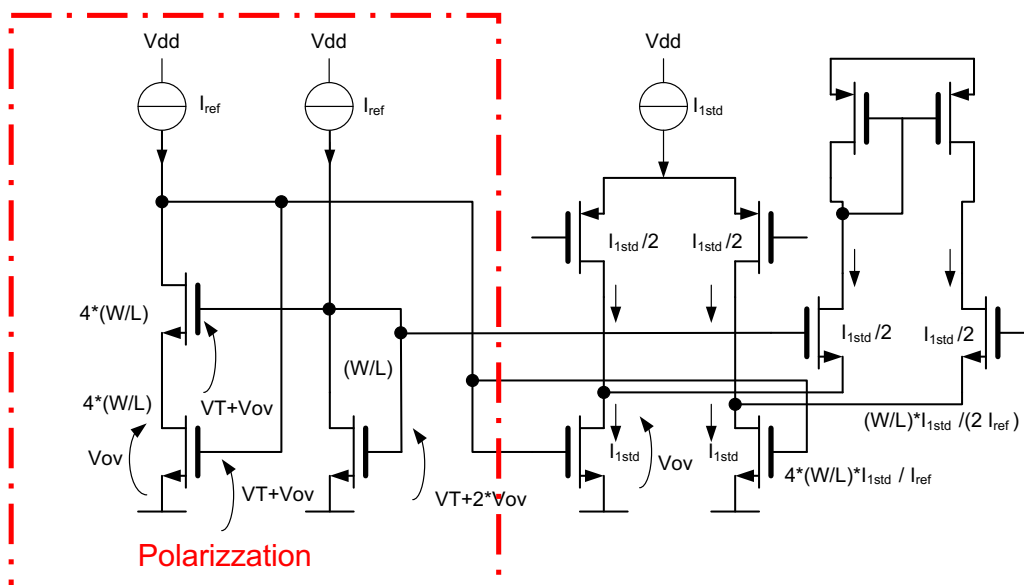


Low Noise design

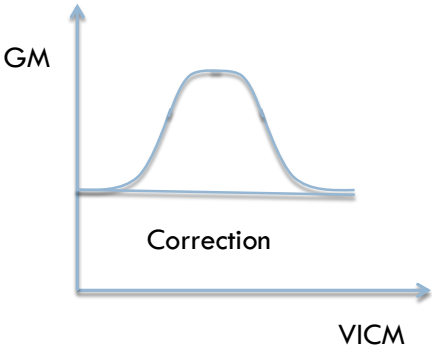
- The PMOS have about 2 to 5 less $1/f$ noise than n
 - ▣ Design a PMOS input stage
- Make the first gain stage as high as possible
 - ▣ Cascode structure to increase the low frequency gain of the opamp

P-ch Folded cascode input schematic

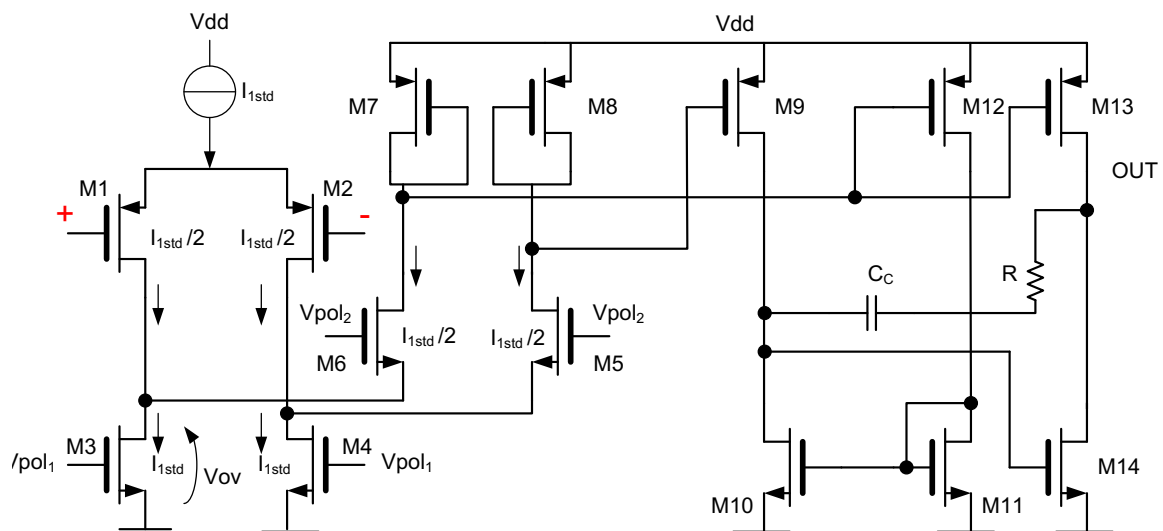
- Increase the Input common mode range



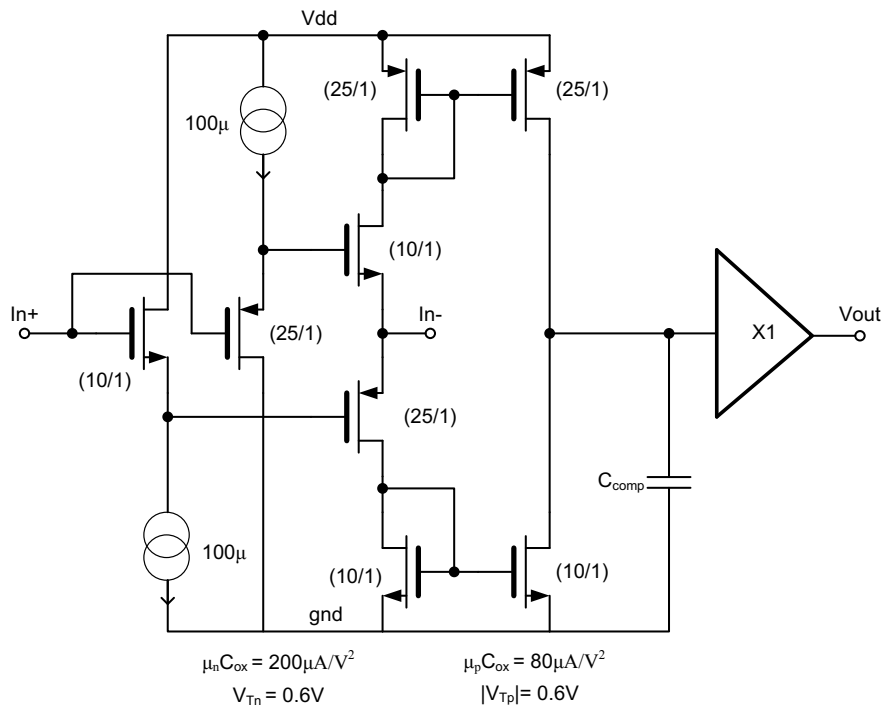
A horizontal bar chart with two bars. The first bar is blue and the second bar is orange. The blue bar is significantly longer than the orange bar.



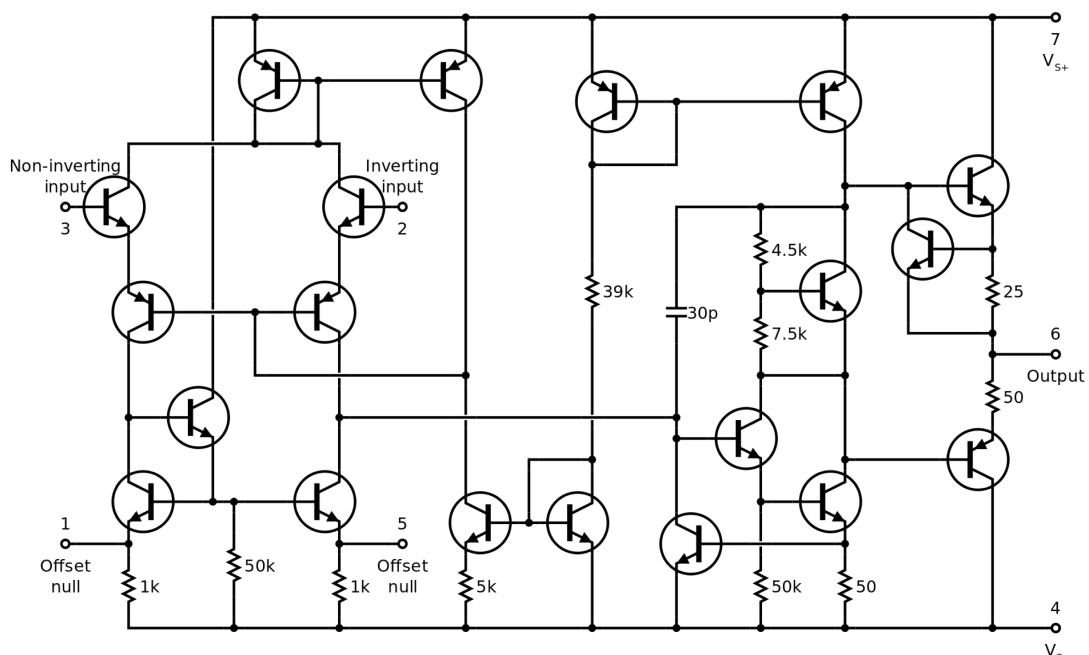
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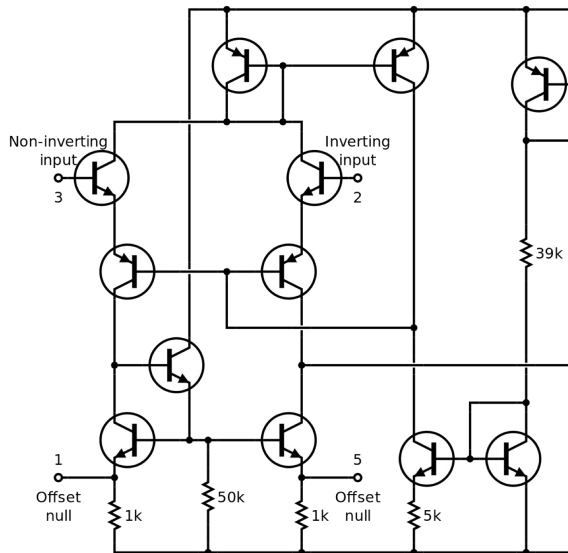
Current feedback



$\mu 741$

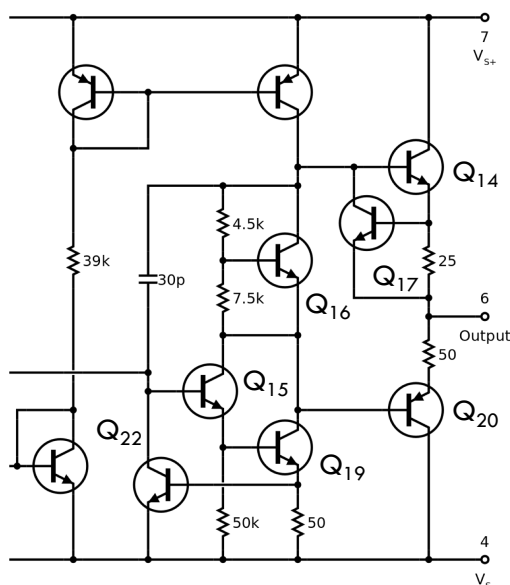


Input stage



- Input stage *npn* higher β
- Gain $g_m/4$ of differential stage gain final $g_m/2$
- Mirror with offset compensation
- Second stage *npn* higher performance
- Bias loop for first stage

Second stage



- Darlington Q15-Q19
- Protection Q22 Q17
- Q14 and Q20 class AB stage
- Q16 v_{be} multiplier ($V_{be} \cdot (1 + 4.7/7.5)$)